FinzyX- an AI-powered Financial Advisor and Analyst



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Chapter 1

1. Introduction

Pakistan's capital markets present one of the most significant untapped financial opportunities in South Asia. The Pakistan Stock Exchange (PSX) hosts over 527 listed companies with a total market capitalization exceeding USD 51.3 billion, yet retail participation remains critically low at 0.13%. Approximately 300,000 active investors against a population of 259 million. This disparity is driven by three persistent structural barriers: cognitive overload from complex financial data, a language barrier that excludes Urdu speaking users and the lack of financial literacy and education among the common population.

Pakistan has 117 million internet users as of 2026 [1], with internet penetration growing rapidly among a population where 64% are below the age of 30. The infrastructure for a web based financial platform is clearly in place. What has been missing is a tool simple enough for the first time investor to actually use. To address this gap, the project team proposes FinzyX: a web based intelligence layer that combines a bilingual AI advisor that acts as an analyst creating a personalized financial roadmap for every user, a user friendly stock screener that does not scare off a first time investor, and a portfolio management module which is a unified dashboard for all types of assets.

1.1. Project Introduction

1.1.1. Brief Description of the Project

The goal of this project is to help the everyday Pakistani retail investor make informed, confident investment decisions on the PSX without requiring a finance degree or fluency in

financial English. FinzyX is built on three integrated modules that together form a complete intelligence layer over Pakistan's capital markets.

AI Rahbar (The Intelligent Advisor): The core engine of FinzyX is an agentic Retrieval Augmented Generation (RAG) system that functions as a 24/7 bilingual financial analyst. The user can type in their query and they will get answered in simpler terms. Users provide their age, monthly income, and risk appetite (Low / Medium / High) according to their personal goals and the AI Rahbar generates a tailored Financial Roadmap suggesting specific asset allocations such as '60% Growth Stocks, 40% Dividends' grounded in real PSX data. It reads and processes complex PSX Annual Reports automatically, interacting seamlessly in English or Roman Urdu so that users never need to navigate a 200 page filing themselves.

Smart Stock Screener (The Discovery Engine): A simplified, beginner friendly tool that replaces the complex filter systems found on platforms like the PSX portal and Sarmaaya. Instead of navigating technical dropdowns and financial ratios, users simply type what they are looking for in plain English or Roman Urdu. For example, "high dividend paying stocks in the banking sector" or "low risk Shariah compliant stocks" and the screener returns a clean, easy to understand list of results tailored to their query.

Portfolio Management: Moving beyond standard profit and loss tracking, this module allows users to log trades manually by entering ticker, price, quantity, and date, with current profit and loss calculated automatically from live PSX prices. A Dividend Calendar displays upcoming payout dates and estimated PKR amounts in a visual timeline, enabling users to plan monthly expenses around expected passive income. It acts as a single dashboard that keeps track of all your assets (stocks, crypto, gold, etc.) in real time.

1.1.2. Introduction of the Beneficiary of the Project

The primary beneficiaries of FinzyX are retail investors and savings oriented individuals in Pakistan who are currently excluded from, or underserved by, the existing PSX investment ecosystem.

- **First time investors and young professionals:** Users such as recent graduates or early career professionals who have savings but lack the financial literacy or confidence to begin investing. FinzyX's personalized AI Rahbar roadmap and simplified filters lower the barrier to entry substantially.
- **Values aligned investors:** Muslim investors who require Shariah compliant investment options. The dedicated Shariah filter, sourced from official certification bodies, allows this demographic to build portfolios consistent with their faith without manual verification.
- **Goal oriented savers:** Users who want a concrete, personalized plan linking their income and risk profile to specific PSX investment recommendations. The AI Rahbar's roadmap output provides a structured starting point tailored to each individual's financial situation.
- **Urdu speaking users:** A large segment of Pakistan's population whose effective exclusion from financial markets is in part due to the English only nature of most financial platforms. FinzyX's Roman Urdu interface directly addresses this barrier.

1.2. Existing Examples / Solutions

This section examines four platforms that Pakistani retail investors currently use or encounter when trying to engage with the PSX. Each platform is evaluated on what it actually does, who it is built for, and critically where it falls short in ways that directly motivated the design of FinzyX.

1.2.1. PSX Official Data Portal (dps.psx.com.pk)

- **Background:** The Data Portal is the official market information system maintained by the Pakistan Stock Exchange. It has been the primary public facing interface for PSX market data since the exchange's demutualization and has been progressively expanded to include a stock screener, financial reports, and a basic portfolio watchlist.

- **Working:** The portal provides live and historical price data for all listed securities, a sector summary, a stock screener with numerical filters (P/E ratio, market cap, volume ranges), downloadable financial reports, corporate announcements, and an AGM/dividend payout calendar. It requires no login for most data and is freely accessible.

Shortcomings:

- The stock screener operates entirely on numerical thresholds. Users must know what a P/E ratio or EPS means before they can use it. There is no way to express intent in plain language, which immediately excludes anyone without a finance background.
- Financial reports are listed as raw PDF downloads. The portal makes no attempt to summarize, translate, or interpret these documents, effectively making them inaccessible to the average investor who cannot parse a 200 page annual report.
- The interface is English only and optimised for data consumption rather than decision support. There is no contextual guidance, no explanation of what the numbers mean, and no mechanism for a user to ask 'what should I do with this information' [2].

1.2.2. Sarmaaya Financials (sarmaaya.pk)

- **Background:** Sarmaaya is a PSX authorized data redistributor and financial research platform launched to serve retail investors with portfolio tracking, stock analytics, and financial education. It has grown into one of the most feature rich third party PSX platforms and has a TradingView powered charting integration.
- **Working:** Users can track a PSX portfolio, screen stocks using fundamental filters such as dividend yield, P/E, and sector, view historical financials, compare multiple stocks, and access a library of educational articles and paid training courses. The platform also covers mutual funds and forex alongside PSX equities.

Shortcomings:

- Despite its breadth, Sarmaaya's screener is filter based and targets users who already understand financial metrics. A first time investor landing on the screener page faces the same wall of numerical inputs as the PSX portal sector dropdowns, ratio ranges, and market cap sliders with no plain language alternative.
- The platform's own App Store reviews acknowledge that financial information is difficult for everyday users to understand, with users explicitly requesting simpler explanations. Sarmaaya's response has been to add more data and paid courses rather than simplify the core experience.
- All content is in English. There is no Urdu or Roman Urdu support at any level of the platform, which structurally limits its reach to the English literate segment of potential investors [3].

1.2.3. Investify (investify.pk)

- **Background:** Investify is a Pakistani fintech platform providing PSX stock data, charts, and a portfolio tracker. It positions itself as a tool for the 'informed retail investor' and includes a Shariah compliance filter as one of its screener options.
- **Working:** Users can browse all PSX listed companies, view price charts and key financial ratios, apply a Shariah compliance toggle to the screener, and maintain a personal watchlist. The interface is cleaner than the PSX portal but follows the same filter driven discovery model.

Shortcomings:

- Investify's Shariah filter is its most differentiating feature, yet it functions purely as a binary toggle. It tells users which stocks are compliant but provides no guidance on which of those compliant stocks are actually worth considering for someone with a specific income level or risk appetite.
- Stock discovery still requires the user to browse company by company or apply predefined numerical filters. The platform assumes users come with a stock in mind

rather than a question, which is the opposite of how a first time investor approaches the market.

- There is no personalization layer whatsoever. Every user sees the same interface regardless of whether they are a 22 year old salaried employee with PKR 10,000 to invest or a 45 year old business owner planning for retirement [4].

1.2.4. Emlaak Financials (emlaakfinancials.com)

- **Background:** Emlaak Financials is a CDC (Central Depository Company) subsidiary launched in 2021 and formally inaugurated in 2022 as Pakistan's first digital mutual fund aggregator. It was built with an explicit mandate to increase the mutual fund penetration rate in Pakistan, which stood at just 1.6% of GDP at the time of launch.
- **Working:** The platform aggregates mutual fund products from multiple Asset Management Companies (AMCs) onto a single portal, allowing users to compare funds, open a Sahulat or Sarmayakari account digitally, and invest, redeem, or convert holdings without visiting an AMC office. It offers consolidated portfolio tracking and return calculators, and has a mini-app integration with EasyPaisa for micro-investments.

Shortcomings:

- Emlaak is exclusively a mutual fund platform. It has no access to PSX equities, no stock screener, and no mechanism for a user interested in directly owning shares of a listed company. An investor who wants exposure to, say, banking sector stocks cannot do so through Emlaak at all.
- While the platform is more accessible than the PSX portal, fund selection still requires users to compare NAVs, returns, and risk categories across AMCs metrics that carry the same jargon problem as equity data. The educational resources are limited to YouTube tutorial videos rather than contextual, in platform guidance.
- The platform was designed for low risk, first time mutual fund investors and has not evolved to serve users seeking direct equity market participation, personalized investment roadmaps, or any form of conversational financial guidance [5].

1.2.5. State of the Art Comparison

Table 1-3 consolidates the key differentiating features across all four platforms and FinzyX. The comparison is focused on the five capabilities that most directly address the barriers identified in the problem statement: language accessibility, personalized guidance, portfolio management, plain language stock search, and Shariah screening.

Table 1-3: State-of-the-Art Comparison of Pakistani Investment Platforms

Feature / Criterion	PSX Data Portal(dps.psx.com.pk)	Sarmaaya(sarmaaya.pk)	Investify(investify.pk)	EmlaakFinancials	FinzyX(Proposed)
AI Generated Personalized Financial Roadmap	✗	✗	✗	✗	✓
Bilingual Roman Urdu Term Conversion	✗	✗	✗	✗	✓
Shariah Compliance Filter	✗	✓	✓	~	✓
Plain Language (NLP) Stock Search	✗	✗	✗	✗	✓
Portfolio Management	~	✓	✓	✓	✓

Legend: ✓ Full Support ✗ Not Supported ~ Partial / Limited Support

The comparison reveals a clear pattern: the PSX portal, Sarmaaya, and Investify all provide data access and filter based screening, but none simplify the discovery experience for a user who does not already speak the language of finance. Emlaak goes furthest in accessibility but is confined entirely to the mutual fund space. The three capabilities that no existing platform offers simultaneously a bilingual interface, an AI generated personalized roadmap and plain language stock search are precisely the three pillars on which FinzyX is built.

1.3 Business Scope

FinzyX is an AI-powered financial advisory platform built for Pakistan's retail investors, operating as a non-brokerage intelligence layer across WealthTech, FinTech, and EdTech sectors. It targets three key segments: young professionals aged 18–35, Shariah-conscious investors, and first-time investors.

Problem	Solution	Unique Value Proposition	Unfair Advantage	Customer Segments
1. Cognitive overload from PSX data. 2. Language barrier excludes Urdu users. 3. No personalized investment roadmap for retail users.	1. Bilingual AI Rahbar with RAG roadmap. 2. Smart Stock screener. 3. Dividend portfolio tracker.	"FinzyX gives every Pakistani a personalized PSX investment roadmap in their own language."	PSX specific RAG corpus. Bilingual Roman Urdu NLP. Non brokerage advisory model.	Young professionals. Shariah conscious investors. First time retail investors.
Existing Alternatives	**Key Metrics**	**High-Level Concept**	**Channels**	**Early Adopters**
PSX portal (data only). Sarmaaya/Investify (screeners). No integrated AI advisory layer.	Monthly active users. AI roadmap completions. Subscription conversion rate.	"FinzyX is like a personal financial advisor for PSX bilingual, AI powered, and built for Pakistan."	Tech Blog University finance societies. Social media (Instagram, X).	University students. Freelancers with savings. Young salaried professionals aged 22–30.
Cost Structure		**Revenue Structure**		
LLM API fees. Pinecone hosting. PSX data feed.		First month free trial. Premium subscription (PKR 3500/month onwards).		

Table 2 Leman Canvas

1.4. Useful Tools and Technologies

This section describes the technologies selected for the design, development, and deployment of FinzyX. Each tool has been evaluated based on its functionality, compatibility with the project requirements, and contribution to the overall system objectives.

1.4.1. Frontend: Next.js and React

The frontend of FinzyX will be built using Next.js 16 with React Server Components, providing a fast, responsive web application experience without requiring any installation from the user. Tailwind CSS 4 combined with Shadcn/UI will handle styling, ensuring a clean and accessible interface suitable for first time investors. TanStack Query will manage server state and data fetching, keeping the UI in sync with live PSX data. Data visualization across the Portfolio and Screener modules will be handled by Recharts or Tremor, enabling intuitive charts and financial summaries.

1.4.2. Backend: Python and FastAPI

The backend will be developed using Python with FastAPI, chosen for its high performance, automatic API documentation, and native compatibility with Python's AI and data science ecosystem. FastAPI will expose RESTful endpoints consumed by the Next.js frontend, handling all business logic including portfolio computation, PSX data retrieval, and query routing to the AI layer. The backend will be deployed on a serverless infrastructure using AWS Lambda or Vercel, ensuring scalability without the overhead of managing dedicated servers.

1.4.3. Database: PostgreSQL

PostgreSQL will serve as the primary relational database, hosted and managed through Supabase. It will store all user specific data including profiles, age and income inputs for roadmap generation, portfolio holdings, trade history, and dividend records. Supabase also provides built in authentication with multi factor authentication support via Supabase Auth or Clerk, removing the need to build a separate identity management system from scratch.

1.4.4. Vector Database: Pinecone

Pinecone will be used as the vector database powering the RAG pipeline of the AI Rahbar. PSX annual reports, regulatory filings, and curated financial news will be chunked, embedded, and indexed in Pinecone. At query time, the user's input will be converted to a vector embedding and matched against the indexed corpus to retrieve the most contextually relevant document chunks, which are then passed to the language model as a grounded context for generating the personalized financial roadmap.

1.4.5. AI Orchestration: LangGraph

The AI Rahbar will be orchestrated using LangGraph or CrewAI, which provide agentic workflow management allowing the system to chain retrieval, reasoning, and response generation steps in a structured and controllable pipeline. PydanticAI will be used for validating all structured inputs and outputs within the agent, ensuring that user profile data such as age, income, and risk appetite is correctly typed and handled before being passed into the model. The underlying language model will be Gemini 3.1 Pro or GPT-4o, accessed via API.

1.4.6. Development Environment

Visual Studio Code will serve as the primary development environment across the team. Git with GitHub will be used for version control and collaborative development, with branch protection rules enforced on the main codebase. Docker will be used to containerize local development environments, ensuring consistency across all team members' machines regardless of operating system.

1.4.7. Supported Platforms

FinzyX will be developed as a web application accessible through any modern browser on desktop or mobile, eliminating the need for users to download or install anything. The frontend will be built with Next.js, ensuring fast page loads through server-side rendering and straightforward deployment on cloud infrastructure. All communication between the client and server will be encrypted via HTTPS, ensuring data security for all user financial information.

1.5. Project Work Breakdown

All projects require careful planning, including a clear outline of task allocation across team members. The project work breakdown divides the FinzyX system into manageable phases and assigns responsibilities based on each member's technical strengths, ensuring effective delivery within the academic timeline.

Team Members and Roles

- **Abdullah (Member 1) AI Engineer:** LangGraph orchestration, RAG pipeline design and Pinecone indexing.
- **Hassaan (Member 2) Backend Engineer:** FastAPI development, PostgreSQL and PSX data integration.
- **Abdullah (Member 3) Frontend Engineer:** Next.js 16 application, Tailwind CSS and Shadcn/UI implementation.

Ph ase	Tasks / Activities	Assigned Member(s)	Duration	Justification
1	Requirement Analysis & Research	All Members	Weeks 1–2	Shared phase ensures alignment on project scope.
2	Literature Review & Feasibility Study	All Members	Weeks 3–4	Validates tech stack before development begins.
3	System Architecture Design	Hassaan & Abdullah (AI)	Weeks 5–7	Backend and AI module coordination is critical.
4	UI/UX Design & Wireframing	Abdullah (Frontend)	Weeks 8–10	Early wireframes validate UX before backend coupling.
5	Backend Setup: APIs, PostgreSQL, PSX Integration	Hassaan	Weeks 11–14	The core data layer must precede AI integration.
6	RAG Pipeline & AI Rahbar Roadmap Engine	Abdullah (AI)	Weeks 15–19	RAG indexing and personalized roadmap logic require dedicated time.
7	Bilingual NLP Layer	Abdullah (AI) & Hassaan	Weeks 20–22	Language routing requires backend coordination.
8	Smart Screener: NLP Stock Search	Hassaan & Abdullah (Frontend)	Weeks 23–26	Combines backend scoring with frontend discovery UI.
9	Portfolio Module & Dividend Calendar	Hassaan & Abdullah (Frontend)	Weeks 27–29	Calendar visualization is frontend-led.
10	Full System Integration & Testing	All Members	Weeks 30–33	Joint effort ensures all modules communicate correctly.
11	Optimization, Deployment & Documentation	All Members	Weeks 34–38	Cloud deployment, performance tuning, and documentation.

Table 3 Task Allocation

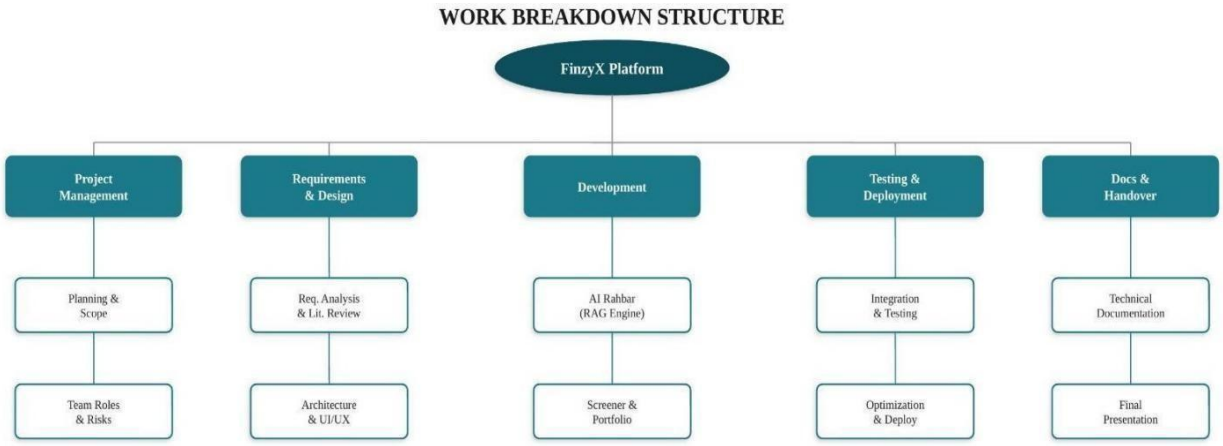


Figure 1-1 Work Breakdown Structure (WBS)

1.6. Project Timeline

The project spans two academic semesters. Semester 7 commenced on 2 March 2026 and covers requirements analysis, literature review, system architecture, UI/UX design, backend setup, and the RAG pipeline core. Semester 8 runs through to January 2027 and covers the bilingual NLP layer, Smart Screener, Portfolio module, full system integration, testing, optimization, deployment, and final presentation.

Figure 1-2 illustrates the project timeline as a Gantt chart. Seven key phases are mapped across the full duration from March 2026 to January 2027, with milestone dates marked at regular intervals.

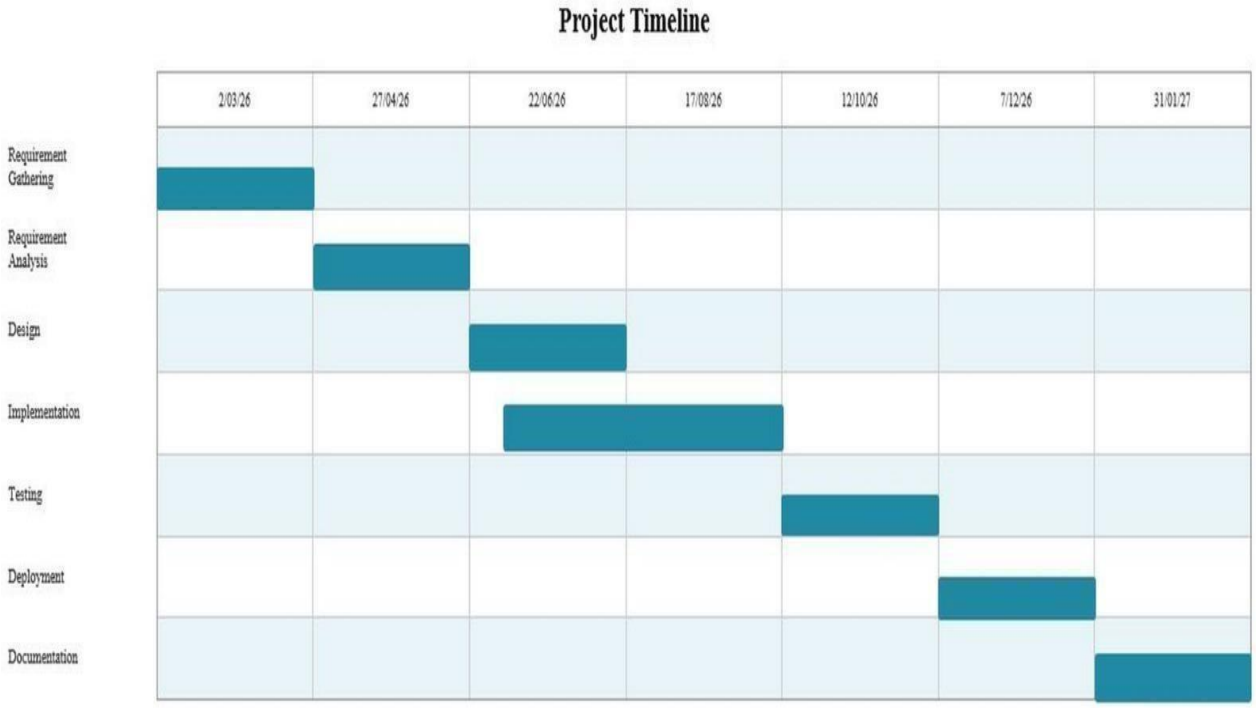


Figure 1-2 Project Gantt Chart

Chapter 2

2. Requirement Specification and Analysis

This chapter identifies and documents the requirements for FinzyX, an AI-powered financial advisory and stock analysis platform built for Pakistani retail investors. The focus is on understanding what the system needs to do and how users will interact with it. Requirements were gathered by studying the gaps in existing Pakistani investment platforms, analysing the financial literacy barriers faced by first-time investors, and mapping out the core workflows that FinzyX must support.

The chapter defines the system's functionalities through epics and user stories. Each epic covers a broad functional area of the platform, while each user story breaks that area into specific, actionable requirements written from the user's perspective. Test cases follow each set of user stories to define the conditions under which each requirement is considered correctly implemented. The chapter concludes with a traceability matrix that maps every epic to its user stories, test cases, and corresponding UI references.

2.1 Epics

Epics define the major functional areas of FinzyX. The system has six epics covering user authentication, the AI Rahbar advisory engine, the natural language stock screener, portfolio management, Shariah compliance filtering, and the bilingual Roman Urdu interface. Together these epics cover everything a first-time Pakistani retail investor needs to go from signing up to getting a personalised investment roadmap and tracking their holdings.

2.1.1 Epic E 1

Title: User Authentication and Onboarding

Description: As a new user, I want to register and log in to FinzyX securely so that my portfolio data, roadmap preferences, and browsing history are private and tied to my account.

2.1.2 Epic E 2

Title: AI Rahbar: Personalised Financial Roadmap

Description: As a registered user, I want to enter my age, monthly income, and risk appetite so that the AI Rahbar analyses my profile against real PSX data and returns a tailored financial roadmap telling me which asset classes to consider and in what proportion.

2.1.3 Epic E 3

Title: Smart Stock Screener with Natural Language Search

Description: As a user, I want to type a plain-language query such as 'high dividend paying stocks in the banking sector' and receive a clean, ranked list of matching PSX stocks so that I can discover investment options without needing to understand financial ratios or use complex filter dropdowns.

2.1.4 Epic E 4

Title: Portfolio Management and Dividend Calendar

Description: As a user, I want to log my stock trades manually and see my current profit or loss calculated from live PSX prices, along with a calendar showing upcoming dividend payout dates and estimated PKR amounts, so that I have a single dashboard for all my holdings.

2.1.5 Epic E 5

Title: Shariah Compliance Filtering

Description: As a Muslim investor, I want to toggle a Shariah compliance filter so that every stock result across the screener and AI roadmap only includes PSX companies that are certified as Shariah compliant, without me having to verify each one manually.

2.1.6 Epic E 6

Title: Bilingual Roman Urdu Interface

Description: As a user who is more comfortable writing in Roman Urdu than English, I want to type my queries and receive responses in Roman Urdu so that the language barrier does not stop me from using the platform.

2.1.7 Epic E 7

Title: Virtual Trading & Learning Environment

Description: As a beginner investor, I want to receive a virtual balance when I create an account so that I can practice investing in Pakistan Stock Exchange (PSX) stocks without risking real money and learn how profits and losses are calculated.

2.2 User Stories

User stories describe FinzyX's features from the perspective of the people using the platform. Each story captures a concrete need, explains why that need matters, and sets out the acceptance criteria that confirm the feature is working as intended. The stories below cover all six epics and are written to reflect the range of users FinzyX is built for, from first-time investors entering the platform for the first time to goal-oriented savers checking their dividend calendar.

2.2.1 E1- US 1

Title: User Registration

Description: As a new user, I want to create an account with my email and a password so that I can access the platform and save my data.

Acceptance Criteria: Given the registration page, when the user enters a valid email address and a password that meets the minimum strength requirements and submits the form, then the system creates the account, sends a verification email, and redirects the user to the onboarding page.

2.2.2 E1- US 2

Title: User Login

Description: As a registered user, I want to log in with my email and password so that I can access my portfolio and saved preferences.

Acceptance Criteria: Given the login page, when the user enters the correct email and password, then the system authenticates the credentials and redirects the user to their dashboard.

2.2.3 E1- US 3

Title: Password Recovery

Description: As a user who has forgotten their password, I want a reset link sent to my email so that I can regain access to my account.

Acceptance Criteria: Given the login page, when the user clicks 'Forgot Password' and enters their registered email address, then the system sends a password reset link to that address within two minutes.

2.2.4 E1- US 4

Title: Onboarding Profile Setup

Description: As a new user, I want to enter my age, monthly income, and risk appetite during onboarding so that the AI Rahbar has the inputs it needs to generate my financial roadmap.

Acceptance Criteria: Given the onboarding screen, when the user fills in their age, monthly income, and selects a risk level from Low, Medium, or High, then the system saves the profile and enables the AI Rahbar roadmap feature.

2.2.5 E2- US 1

Title: Generate AI Financial Roadmap

Description: As a user, I want the AI Rahbar to read my profile and return a personalised investment roadmap so that I know exactly what kind of stocks to look at given my income and risk level.

Acceptance Criteria: Given the AI Rahbar page, when the user requests a roadmap, then the system retrieves the user profile, queries the RAG pipeline against indexed PSX data, and returns a roadmap within ten seconds that includes a recommended asset allocation and at least three stock categories relevant to the user's risk level.

2.2.6 E2- US 2

Title: Ask a Financial Question in Plain Language

Description: As a user, I want to type any financial question into the AI Rahbar chat and get a clear, simple answer grounded in PSX data so that I do not have to read raw annual reports myself.

Acceptance Criteria: Given the AI Rahbar chat interface, when the user types a question about a PSX stock or financial concept, then the system returns a response in plain language within ten seconds that references at least one PSX source document.

2.2.7 E2- US 3

Title: Update Profile and Regenerate Roadmap

Description: As a user whose financial situation has changed, I want to update my income or risk level so that the AI Rahbar generates a revised roadmap reflecting my current position.

Acceptance Criteria: Given the profile settings page, when the user updates any of the three profile fields and saves the changes, then the AI Rahbar roadmap on the next generation reflects the updated inputs.

2.2.8 E2- US 4

Title: View Financial News Feed

Description: As a user, I want to view the latest financial and stock market news,

So that I can make informed investment decisions based on current market events

Acceptance Criteria: Given the user opens the News section, when the application requests the latest news from the integrated financial news API, then the system displays recent financial news articles including headlines, summaries, publication dates, and source information.

2.2.9 E3- US 1

Title: Search Stocks Using Natural Language

Description: As a user, I want to type what I am looking for in plain English or Roman Urdu and get a ranked list of matching PSX stocks so that I do not have to use filter sliders or know what a P/E ratio is.

Acceptance Criteria: Given the screener search bar, when the user types a natural language query such as 'low risk cement stocks' and submits it, then the system returns a list of at least three relevant PSX stocks ranked by match quality within five seconds.

2.2.10 E3- US 2

Title: View Stock Detail Page

Description: As a user who found a stock in the screener results, I want to click on it and see key information such as current price, sector, and a brief AI-generated summary so that I can decide whether it is worth investigating further.

Acceptance Criteria: Given a screener results page, when the user clicks on any listed stock, then the system opens a stock detail page showing the current PSX price, sector, a oneparagraph AI-generated company summary, and a button to add it to the portfolio.

2.2.11 E3- US 3

Title: Empty State for Unrecognised Query

Description: As a user who types a query the system cannot match, I want to see a clear message and a suggestion rather than a blank screen so that I am not left confused.

Acceptance Criteria: Given the screener, when the user submits a query that returns no matching stocks, then the system displays a message stating no results were found and suggests two alternative query phrasings.

2.2.12 E4- US 1

Title: Log a Stock Trade Manually

Description: As a user, I want to add a trade by entering a ticker symbol, buy price, quantity, and date so that the portfolio tracks my holdings without me connecting a brokerage account.

Acceptance Criteria: Given the portfolio page, when the user enters a valid PSX ticker, a buy price, a quantity greater than zero, and a date, then the system adds the holding to the portfolio and immediately calculates profit or loss using the current PSX price.

2.2.13 E4- US 2

Title: View Live Portfolio Profit and Loss

Description: As a user, I want to see my overall portfolio value and the gain or loss on each holding updated from live PSX prices so that I know how my investments are performing today.

Acceptance Criteria: Given the portfolio dashboard, when the user opens it during trading hours, then every holding shows the current market price fetched from the PSX data feed, and the total portfolio value and percentage change are recalculated and displayed.

2.2.14 E4- US 3

Title: Dividend Calendar

Description: As a user with dividend-paying stocks in my portfolio, I want to see a calendar of upcoming dividend payout dates and estimated PKR amounts so that I can plan around expected income.

Acceptance Criteria: Given the dividend calendar view, when the user opens it, then the system displays the next three months of dividend events sourced from PSX corporate announcements, each showing the company ticker, ex-dividend date, and an estimated payout amount in PKR based on the user's holdings.

2.2.15 E4- US 4

Title: Edit an Existing Trade

Description: As a user, I want to edit the price or quantity of a trade I have already logged so that I can correct mistakes without having to delete and re-enter the full entry.

Acceptance Criteria: Given the portfolio page, when the user selects an existing trade, updates the buy price or quantity to a valid value, and saves the change, then the system updates the holding record and immediately recalculates the profit or loss using the corrected values and the current PSX price.

2.2.16 E4- US 4

Title: Delete a Trade

Description: As a user, I want to delete a trade I have logged in error so that my portfolio dashboard accurately reflects only my actual holdings.

Acceptance Criteria: Given the portfolio page, when the user selects a trade, confirms the deletion through the confirmation prompt, then the system permanently removes the holding from the portfolio and

2.2.17 E5- US 1

Title: Enable Shariah Compliance Filter

Description: As a Muslim investor, I want to switch on a Shariah filter so that only PSX stocks with a valid Shariah compliance certification appear in my screener results and AI roadmap.

Acceptance Criteria: Given any page with stock results, when the user enables the Shariah compliance toggle, then the system immediately re-filters all displayed stocks and excludes any ticker not present on the official Shariah compliance list, with the toggle state persisting across the session.

2.2.18 E5- US 2

Title: Shariah Status on Stock Detail Page

Description: As a user, I want to see the Shariah compliance status displayed clearly on each stock's detail page so that I do not have to check separately.

Acceptance Criteria: Given a stock detail page, when it loads, then a clearly visible label shows either 'Shariah Compliant' or 'Not Shariah Compliant' sourced from the official certification list

2.2.19 E6- US 1

Title: Submit Query in Roman Urdu

Description: As a user more comfortable writing in Roman Urdu than English, I want to type my screener or AI Rahbar query in Roman Urdu and receive a response in Roman Urdu so that the language gap does not stop me from using the platform.

Acceptance Criteria: Given the screener or AI Rahbar input field, when the user types a query in Roman Urdu, then the system detects the language, processes the query, and returns results with labels and the AI response written in Roman Urdu.

2.2.20 E6- US 2

Title: Language Preference Setting

Description: As a user, I want to set my preferred language to English or Roman Urdu in my account settings so that the platform defaults to that lanlguage every time I log in.

Acceptance Criteria: Given the account settings page, when the user selects English or Roman Urdu as their preferred language and saves, then all subsequent AI Rahbar responses and screener labels appear in the chosen language until the preference is changed.

2.2.21 E7- US 1

Title: Receive Dummy Balance

Description: As a new user, I want to receive PKR 50,000 in virtual balance after signing up, So that I can start learning stock trading immediately.

Acceptance Criteria: Given a new user successfully completes the registration process, when the account is created for the first time, then the system automatically credits the user with PKR 50,000 as a virtual balance and displays it on the dashboard.

2.2.22 E7- US 2

Title: Buy Stocks Using Virtual Balance

Description: As a user, I want to purchase stocks using my virtual balance,
So that I can practice investing without financial risk.

Acceptance Criteria: Given the user has sufficient virtual balance, when the user selects a stock, enters the quantity, and confirms the purchase, then the system deducts the corresponding amount from the virtual balance, adds the stock to the user's portfolio, and updates the portfolio value.

2.2.23 E7- US 3

Title: Sell Stocks

Description: As a user, I want to sell my purchased stocks,
So that I can understand profit and loss calculations.

Acceptance Criteria: Given the user owns stocks in their virtual portfolio, when the user sells part or all of the holdings, then the system calculates the profit or loss based on the current market price, updates the virtual balance, and adjusts the portfolio accordingly.

2.3 Acceptance Criteria: Test Cases

Test cases set out the specific conditions used to verify that each user story is correctly implemented. Each test case names an input, describes the action taken, and states the expected outcome. Two test cases are provided per user story: one for the successful path and one for an error or edge condition. This ensures the system handles both normal use and failure scenarios correctly.

Table 2- 1: Successful User Registration

Test ID: E1-US1-TC1	US ID: E1-US1
Test Case Description	Verify a new user can register successfully with a valid email and a strong password.
Inputs	User enters a valid email address not already in the system and a password of at least eight characters containing a number and a capital letter.
Expected Result	System creates the account, sends a verification email to the provided address, and redirects the user to the onboarding profile page.

Table 2- 2: Registration Fails with Duplicate Email

Test ID: E1-US1-TC2	US ID: E1-US1
Test Case Description	Verify the system rejects registration when the email address is already registered.
Inputs	User enters an email address that already exists in the system database.
Expected Result	System displays an error message stating the email is already in use and the account is not created.

Table 2-3: Successful Login with Correct Credentials

Test ID: E1-US2-TC1	US ID: E1-US2
Test Case Description	Verify a registered user can log in with the correct email and password.
Inputs	User enters a valid registered email address and the correct matching password.
Expected Result	System authenticates the user and redirects to the main dashboard.

Table 2-4: Login Fails with Wrong Password

Test ID: E1-US2-TC2	US ID: E1-US2
Test Case Description	Verify the system denies login when the password is incorrect.
Inputs	User enters a valid registered email address and an incorrect password.
Expected Result	System displays an 'Invalid credentials' error and does not grant access.

Table 2-5: Login Fails with Unregistered Email

Test ID: E1-US2-TC3	US ID: E1-US2
Test Case Description	Verify the system denies login when the email address does not exist in the system.
Inputs	User enters an email address that has no registered account and any password.
Expected Result	System displays a 'No account found with this email address.' error and does not grant access.

Table 2-6: Onboarding Fails with Invalid Age and Income Values

Test ID: E1-US4-TC3	US ID: E1-US4
Test Case Description	Verify the system rejects onboarding input when age or income values are logically invalid.
Inputs	User enters Age as −5 and Monthly Income as 0, then attempts to submit the onboarding form.
Expected Result	System highlights both invalid fields and displays a validation message: 'Please enter a valid age and income.' The profile is not saved and the AI Rahbar roadmap feature remains inaccessible.

Table 2-7: Password Reset Email Sent

Test ID: E1-US3-TC1	US ID: E1-US3
Test Case Description	Verify the system sends a password reset email when a valid registered address is submitted.
Inputs	User clicks 'Forgot Password' and enters their registered email address.
Expected Result	System sends a password reset link to the provided email within two minutes.

Table 2-8: Password Reset Fails for Unregistered Email

Test ID: E1-US3-TC2	US ID: E1-US3
Test Case Description	Verify the system does not send a reset email for an address that has no account.
Inputs	User enters an email address that does not exist in the system.
Expected Result	System displays 'No account found with this email address.'

Table 2-9: Onboarding Profile Saved Successfully

Test ID: E1-US4-TC1	US ID: E1-US4
Test Case Description	Verify the system saves a complete onboarding profile and enables the AI Rahbar.
Inputs	User enters age as 26, monthly income as PKR 80 ,000, and selects 'Medium' risk appetite.
Expected Result	System stores the profile, marks onboarding complete, and the AI Rahbar roadmap feature becomes accessible.

Table 2-10: Onboarding Fails with Missing Fields

Test ID: E1-US4-TC2	US ID: E1-US4
Test Case Description	Verify the system does not proceed when any required onboarding field is empty.
Inputs	User leaves the monthly income field blank and attempts to submit.
Expected Result	System highlights the empty field and displays a validation message asking the user to complete all required fields.

Table 2-11: AI Roadmap Generated for Valid Profile

Test ID: E2-US1-TC1	US ID: E2-US1
Test Case Description	Verify the AI Rahbar returns a personalised roadmap for a user with a complete profile.
Inputs	User with age 28, income PKR 60,000, and Low risk appetite requests a roadmap.
Expected Result	System returns a roadmap within ten seconds that includes an asset allocation recommendation and at least three stock categories appropriate for a low-risk profile.

Table 2-12: Roadmap Blocked for Incomplete Profile

Test ID: E2-US1-TC2	US ID: E2-US1
Test Case Description	Verify the system does not generate a roadmap when the user profile is incomplete.
Inputs	User skips onboarding and navigates directly to the AI Rahbar roadmap page.
Expected Result	System displays a prompt asking the user to complete their profile before a roadmap can be generated.

Table 2-13: AI Rahbar Answers a Financial Question

Test ID: E2-US2-TC1	US ID: E2-US2
Test Case Description	Verify the AI Rahbar returns a relevant, sourcegrounded answer to a plain-language financial question.
Inputs	User types: 'What is the dividend yield of Engro Fertilizers this year?'
Expected Result	System returns a response in plain language within ten seconds that includes a figure sourced from a PSX filing or annual report, with the source document cited.

Table 2-14: AI Rahbar Handles Unanswerable Question Gracefully

Test ID: E2-US2-TC2	US ID: E2-US2
Test Case Description	Verify the system responds helpfully when the query falls outside available PSX data.
Inputs	User types: 'What is the share price of Apple on the New York Stock Exchange?'
Expected Result	System responds that it covers PSX-listed stocks only and suggests the user rephrase the query to focus on a Pakistani market company.

Table 2-15: Profile Update Triggers Revised Roadmap

Test ID: E2-US3-TC1	US ID: E2-US3
Test Case Description	Verify the roadmap updates when the user changes their risk level.
Inputs	User changes risk appetite from Low to High in profile settings and generates a new roadmap.
Expected Result	The new roadmap reflects the updated risk level and shows different stock categories compared to the previous Low-risk roadmap.

Table 2-16: Natural Language Query Returns Relevant Stocks

Test ID: E3-US1-TC1	US ID: E3-US1
Test Case Description	Verify the screener returns relevant PSX stocks for a plain-language query.
Inputs	User types: 'high dividend paying stocks in the banking sector' and submits.
Expected Result	System returns a ranked list of at least three PSX banking sector stocks known for above-average dividend yields within five seconds.

Table 2-17: Roman Urdu Query Returns Results

Test ID: E3-US1-TC2	US ID: E3-US1
Test Case Description	Verify the screener handles a query typed in Roman Urdu.
Inputs	User types: 'halal stocks jo zyada dividend dein' (Shariah compliant high dividend stocks) and submits.
Expected Result	System detects Roman Urdu input, processes the query, and returns a list of Shariah-compliant high dividend PSX stocks.

Table 2-18: Stock Detail Page Loads with All Required Fields

Test ID: E3-US2-TC1	US ID: E3-US2
Test Case Description	Verify the stock detail page shows all required information when a result is clicked.
Inputs	User clicks on 'MCB Bank Limited' from the screener results page.
Expected Result	System opens the MCB detail page showing current PSX price, sector, a one-paragraph AI summary, and an 'Add to Portfolio' button.

Table 2-19: Empty State Message Shown for Unrecognised Query

Test ID: E3-US3-TC1	US ID: E3-US3
Test Case Description	Verify a helpful message is shown when the screener finds no matching stocks.
Inputs	User types a nonsense string that matches no PSX stocks or recognised financial intent.
Expected Result	System displays 'No results found' along with two suggested alternative query phrasings.

Table 2-20: Trade Logged and P&L Calculated

Test ID: E4-US1-TC1	US ID: E4-US1
Test Case Description	Verify the system logs a trade and immediately calculates profit or loss.
Inputs	User enters ticker 'LUCK', buy price PKR 500, quantity 100, and date 01/03/2026.
Expected Result	System adds the holding to the portfolio and displays the current market value alongside a profit or loss figure based on the live PSX price.

Table 2-21: Valid Trade Edit Triggers P&L Recalculation

Test ID: E4-US4-TC1	US ID: E4-US4
Test Case Description	Verify the system updates the holding and recalculates P&L when a valid trade edit is saved.
Inputs	User selects the LUCK trade (buy price PKR 500, quantity 100), changes the buy price to PKR 480 and quantity to 120, then saves.
Expected Result	System updates the LUCK holding with the new values and immediately displays a recalculated profit or loss figure based on PKR 480 buy price, 120 shares, and the current live PSX price.

Table 2-22: Invalid Price Rejected During Trade Edit

Test ID: E4-US4-TC2	US ID: E4-US4
Test Case Description	Verify the system rejects a trade edit when an invalid price is entered.
Inputs	User attempts to edit the buy price of an existing trade to −100 (a negative value).
Expected Result	System highlights the price field and displays a validation error: 'Please enter a valid buy price greater than zero.' The holding is not updated.

Table 2-23: Confirmed Deletion Removes Trade from Portfolio

Test ID: E4-US5-TC1	US ID: E4-US5
Test Case Description	Verify the system permanently removes a trade f from the portfolio when the user confirms the deletion.
Inputs	User selects the ENGRO holding, clicks the delete option, and confirms by pressing 'Delete' on the confirmation prompt.
Expected Result	System removes the ENGRO entry from the portfolio holdings list and updates the total portfolio value and overall P&L to exclude the deleted trade.

Table 2-24: Cancelled Deletion Leaves Trade Unchanged

Test ID: E4-US5-TC2	US ID: E4-US5
Test Case Description	Verify the system does not remove any trade when the user cancels the deletion confirmation.
Inputs	User selects the ENGRO holding, clicks the delete option, and presses 'Cancel' on the confirmation prompt.
Expected Result	System dismisses the prompt and the ENGRO holding remains in the portfolio unchanged. No values are modified.

Table 2-25: Invalid Ticker Rejected

Test ID: E4-US1-TC2	US ID: E4-US1
Test Case Description	Verify the system rejects a trade entry when the ticker does not exist on the PSX.
Inputs	User enters the ticker 'XXXX', which is not listed on the PSX.
Expected Result	System displays 'Ticker not found on PSX. Please check the symbol and try again.' and does not save the entry.

Table 2-26: Portfolio Dashboard Reflects Live Prices

Test ID: E4-US2-TC1	US ID: E4-US2
Test Case Description	Verify the portfolio dashboard updates values using live PSX prices during market hours.
Inputs	User opens the portfolio dashboard while the PSX market is open.
Expected Result	All holdings display current market prices sourced from the PSX feed and the total portfolio value and percentage change are recalculated and visible.

Table 2-27: Portfolio Dashboard Shows Last Known Prices After Market Close

Test ID: E4-US2-TC2	US ID: E4-US2
Test Case Description	Verify the portfolio dashboard displays last known prices and a clear market-closed indicator when the PSX market is not active.
Inputs	User opens the portfolio dashboard outside PSX trading hours (before 9:30 AM or after 3:30 PM PKT).
Expected Result	All holdings display the last recorded closing price and the portfolio header shows a 'Market Closed' badge. No stale or zero prices are displayed.

Table 2-28: Dividend Calendar Shows Upcoming Payouts

Test ID: E4-US3-TC1	US ID: E4-US3
Test Case Description	Verify the dividend calendar shows events for stocks in the user's portfolio.
Inputs	User holds 200 shares of a PSX stock that has an announced dividend with an ex-date within the next 90 days.
Expected Result	The dividend calendar displays the event with the company ticker, ex-dividend date, and an estimated PKR payout calculated from the user's holding quantity.

Table 2-29: No Dividend Events Shown for Portfolios Without Qualifying Holdings

Test ID: E4-US3-TC2	US ID: E4-US3
Test Case Description	Verify the calendar shows a clear empty state when no dividend events apply to the portfolio.
Inputs	User holds stocks that have no announced dividends in the next 90 days.
Expected Result	Calendar displays 'No upcoming dividends in the next 90 days' rather than a blank screen.

Table 2-30: Shariah Filter Removes Non-Compliant Stocks from Results

Test ID: E5-US1-TC1	US ID: E5-US1
Test Case Description	Verify the Shariah filter removes non-compliant stocks from screener results when enabled.
Inputs	User searches for 'banking stocks' with the Shariah compliance toggle switched on.
Expected Result	System returns only PSX banking sector stocks that appear on the official Shariah compliance certification list and excludes all others.

Table 2-31: Filter State Persists Across Screener Searches

Test ID: E5-US1-TC2	US ID: E5-US1
Test Case Description	Verify the Shariah filter stays active when the user submits a second query without toggling it off.
Inputs	User enables the Shariah filter, runs one query, then types a second different query without touching the toggle.
Expected Result	The second set of results also excludes noncompliant stocks and the toggle remains in the enabled state.

Table 2-32: Filter State Persists Across Screener Searches

Test ID: E5-US2-TC2	US ID: E5-US2
Test Case Description	Verify that a red 'Not Shariah Compliant' label is displayed on the detail page of a stock that is not on the official certification list.
Inputs	User opens the detail page of a PSX stock that is not Shariah certified.
Expected Result	Page shows a red 'Not Shariah Compliant' label near the top of the company information section, sourced from the official certification list.

Table 2-33: Shariah Status Displayed on Stock Detail Page

Test ID: E5-US2-TC1	US ID: E5-US2
Test Case Description	Verify the compliance label appears correctly on a stock that is on the Shariah list.
Inputs	User opens the detail page of a PSX stock that is Shariah certified.
Expected Result	Page shows a green 'Shariah Compliant' label near the top of the company information section.

Table 2-34: Roman Urdu Query Processed and Response Returned in Roman Urdu

Test ID: E6-US1-TC1	US ID: E6-US1
Test Case Description	Verify the system detects Roman Urdu input and responds in Roman Urdu.
Inputs	User types 'mujhe banking sector ke ache stocks batao' into the AI Rahbar chat.
Expected Result	System detects Roman Urdu, processes the query against PSX data, and returns a response written in Roman Urdu naming relevant banking sector stocks.

Table 2-35: Mixed-Language Query Processed and Shariah Banking Stocks Returned

Test ID: E6-US1-TC2	US ID: E6-US1
Test Case Description	Verify the system correctly handles a mixed English–Roman Urdu query and returns relevant Shariah-compliant results.
Inputs	User types ‘show me halal banking stocks jo zyada dividend dein’ into the screener or AI Rahbar input field.
Expected Result	System detects the mixed-language input, processes the intent as a Shariah-compliant high-dividend banking sector query, and returns a list of matching PSX stocks with labels and AI response rendered in Roman Urdu.

Table 2-36: Dummy Balance Automatically Assigned After User Registration

Test ID: E7-US1-TC1	US ID: E7-US1
Test Case Description	Verify that a new user automatically receives PKR 50,000 as a dummy balance after successful account registration.
Inputs	User completes the registration form with valid details and creates a new account.
Expected Result	The system successfully creates the user account, automatically credits PKR 50,000 as the initial dummy balance, and displays the balance on the user dashboard.

Table 2-37: Dummy Balance Displayed on Dashboard

Test ID: E7-US1-TC2	US ID: E7-US1
Test Case Description	Verify that the allocated dummy balance is correctly displayed on the user's dashboard after login.
Inputs	User logs into an account that has been assigned the initial dummy balance.
Expected Result	The dashboard displays the available dummy balance of PKR 50,000, allowing the user to begin virtual stock trading.

Table 2-38: Buy Stocks Using Dummy Balance

Test ID: E7-US2-TC1	US ID: E7-US2
Test Case Description	Verify that the user can successfully purchase stocks using the available dummy balance.
Inputs	Verify that the user can successfully purchase stocks using the available dummy balance.
Expected Result	The selected stock is added to the user's virtual portfolio, the purchase amount is deducted from the dummy balance, and the updated portfolio information is displayed successfully.

Table 2-39: Prevent Purchase When Dummy Balance is Insufficient

Test ID: E7-US2-TC2	US ID: E7-US2
Test Case Description	Verify that the system prevents stock purchases when the user does not have sufficient dummy balance.
Inputs	User attempts to purchase stocks whose total value exceeds the available dummy balance.
Expected Result	The transaction is rejected, an appropriate "Insufficient Dummy Balance" message is displayed, and no changes are made to the portfolio or account balance.

Table 2-40:Sell Stocks and Calculate Profit/Loss

Test ID: E7-US3-TC1	US ID: E7-US3
Test Case Description	Verify that the user can successfully sell owned stocks and the system calculates profit or loss correctly.
Inputs	User selects stocks from the virtual portfolio, specifies the quantity to sell, and confirms the transaction.
Expected Result	The system sells the selected shares, calculates the profit or loss using the current stock price, updates the dummy balance, and refreshes the portfolio accordingly.

Table 2-41: Updated Dummy Balance After Selling Stocks

Test ID: E7-US3-TC2	US ID: E7-US3
Test Case Description	Verify that the user's dummy balance and portfolio are updated after completing a stock sale.
Inputs	User successfully completes a stock selling transaction.
Expected Result	The portfolio reflects the updated stock quantity, the dummy balance is adjusted based on the sale value, and the profit or loss information is displayed accurately.

Table 2-42: Financial News Successfully Retrieved

Test ID: E2-US4-TC1	US ID: E2-US4
Test Case Description	Verify that the latest financial news is successfully retrieved from the external news API.
Inputs	Verify that the latest financial news is successfully retrieved from the external news API.
Expected Result	The system retrieves and displays the latest financial news articles with headlines, summaries, publication dates, and source names.

Table 2-43: API Failure Handling

Test ID: E2-US4-TC2	US ID: E2-US4
Test Case Description	Verify that the system handles news API failures gracefully.
Inputs	News API is unavailable or returns an error.
Expected Result	The system displays an appropriate error message such as "Unable to load news at the moment. Please try again later." without affecting other application features.

2.4 User Interface Screens

The following figures present the interface screens corresponding to each user story identified in this chapter. Each screen is referenced in the traceability matrix above. Screenshots will be inserted by the project team prior to the Mid Exam 1 submission.

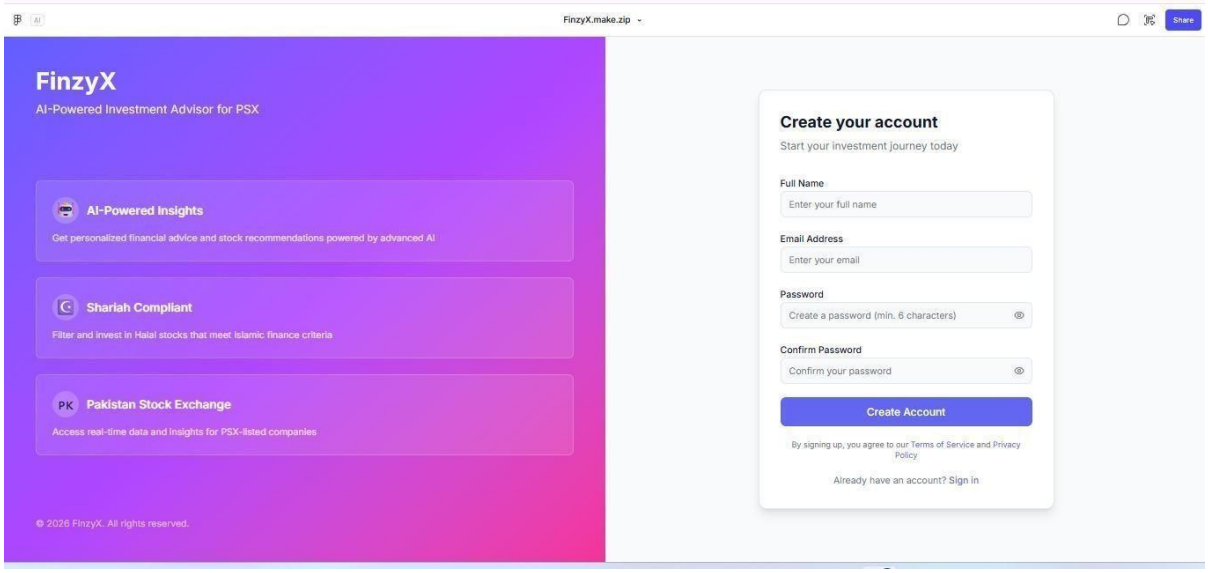


Figure 2-1 Registration Page

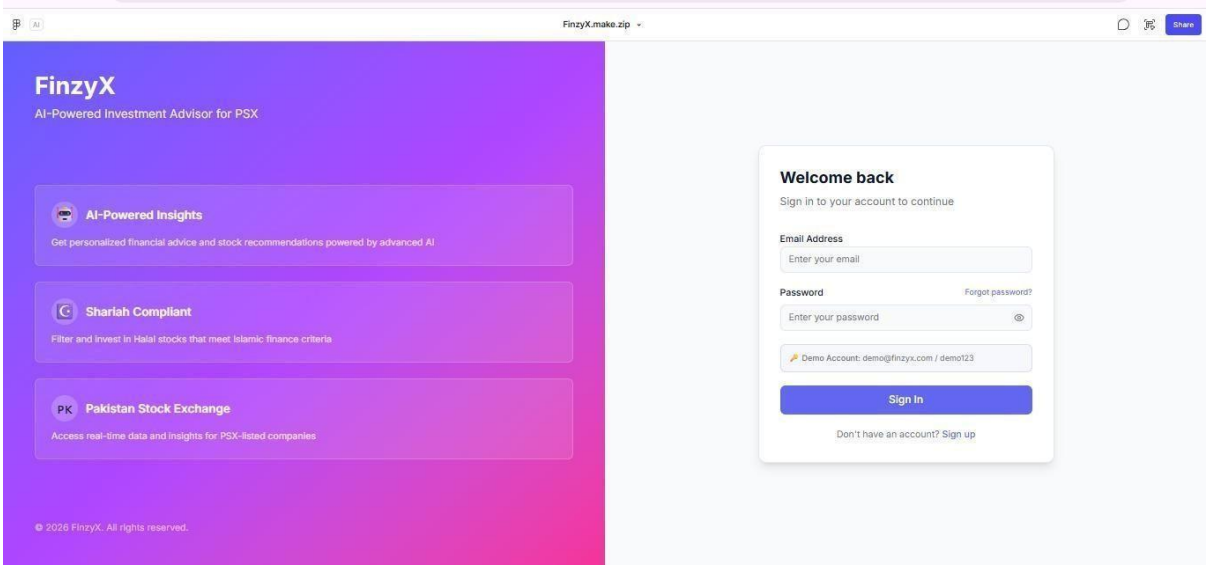


Figure 2-2 Login Page

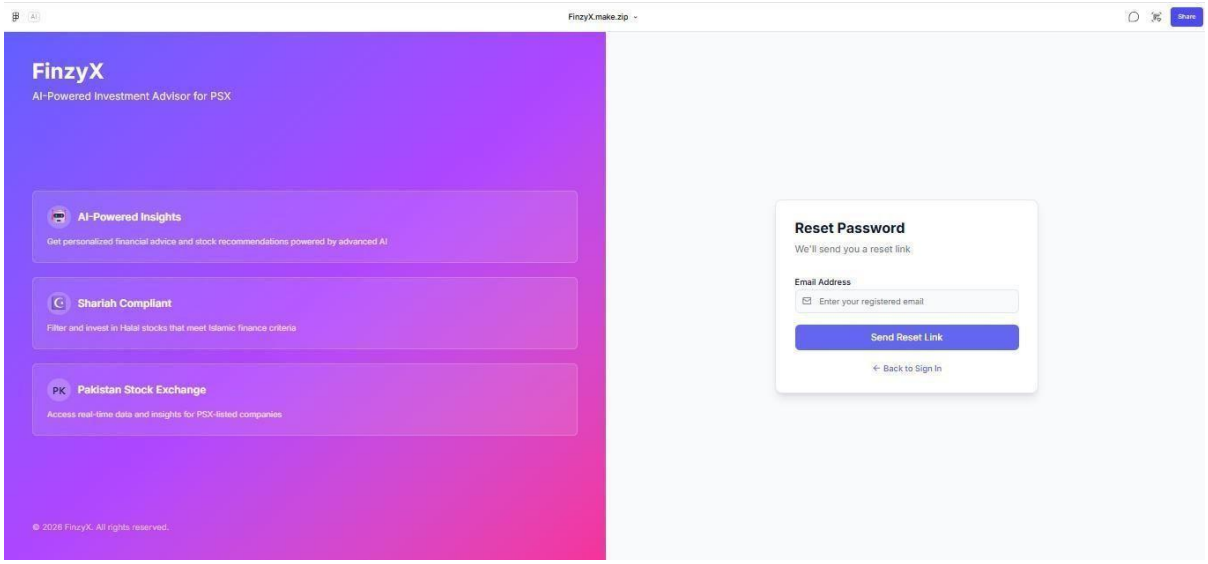


Figure 2-3 Password Recovery Page

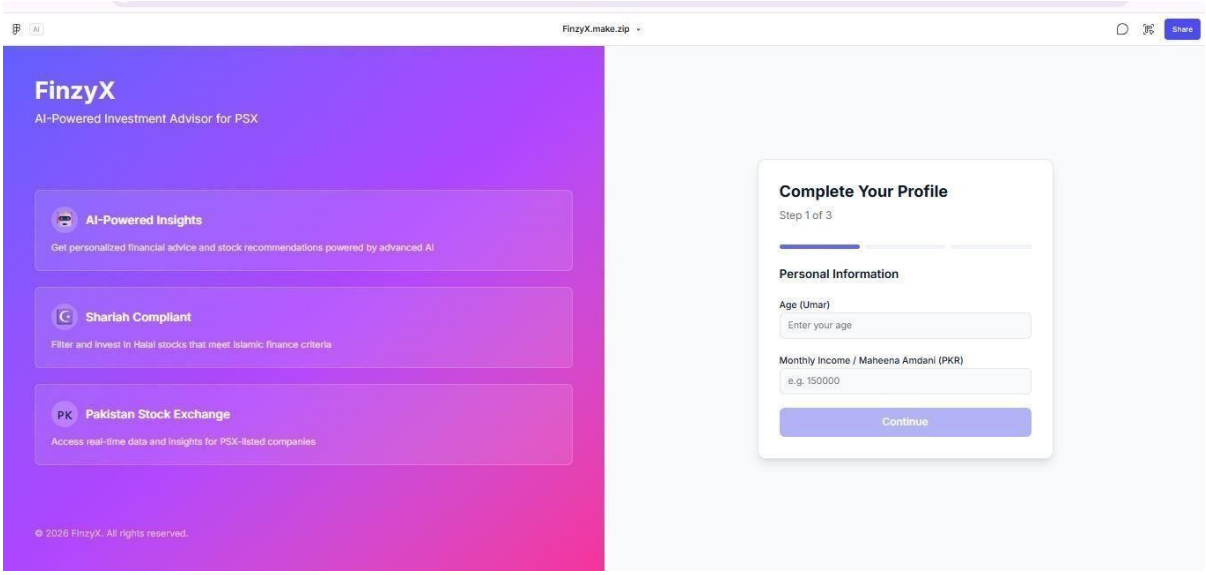


Figure 2-4 Onboarding Profile Setup

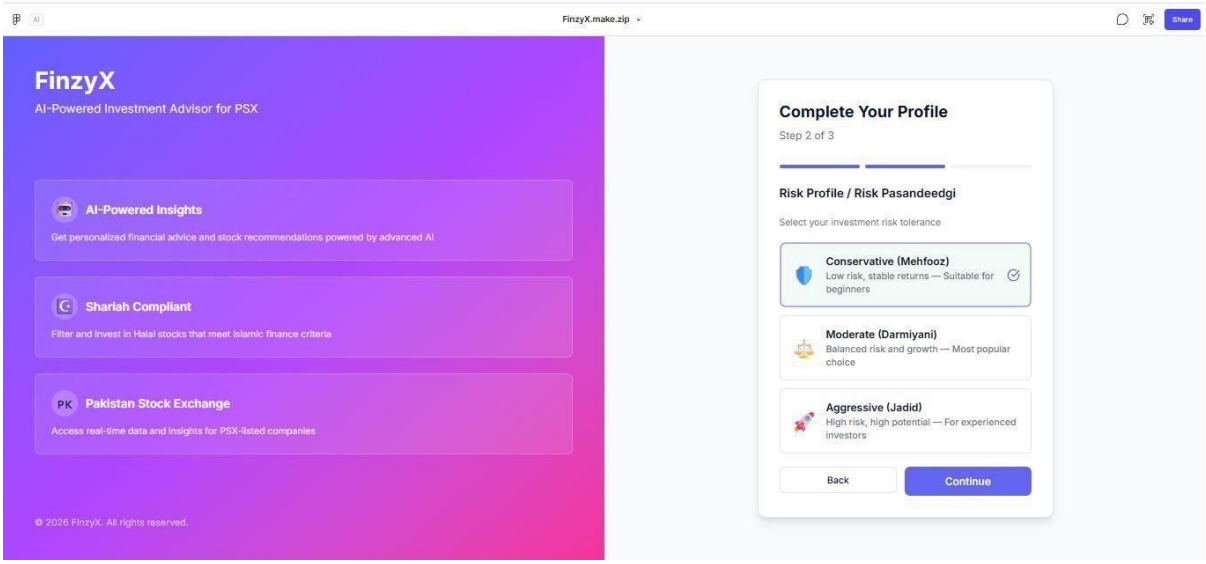


Figure 2-5 Onboarding Profile Setup (Risk)

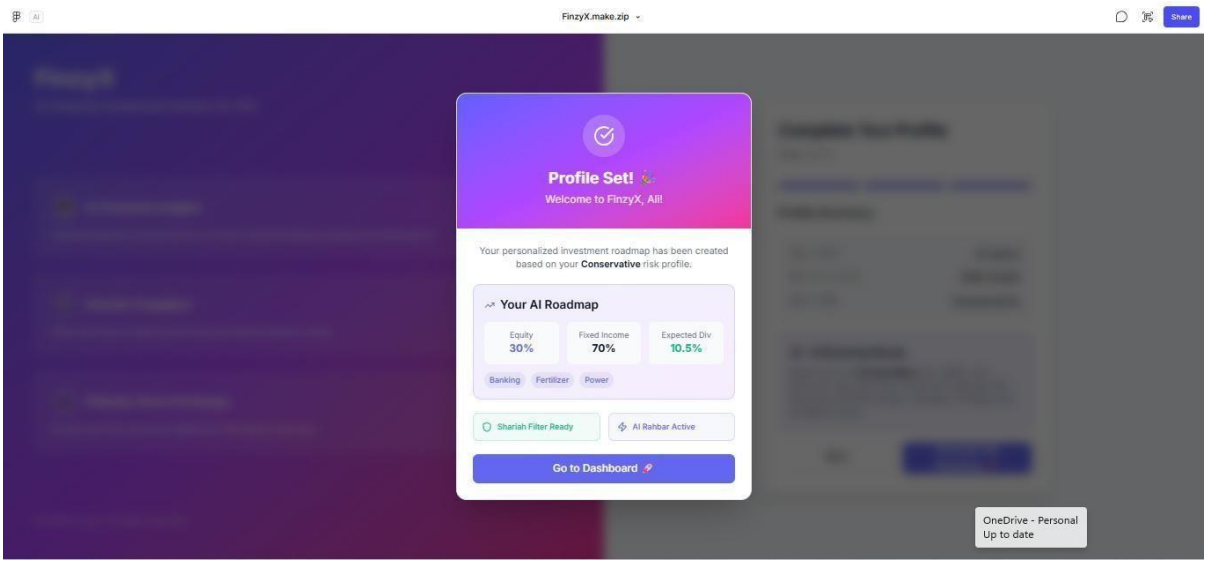


Figure 2-6 AI Rahbar Roadmap Page

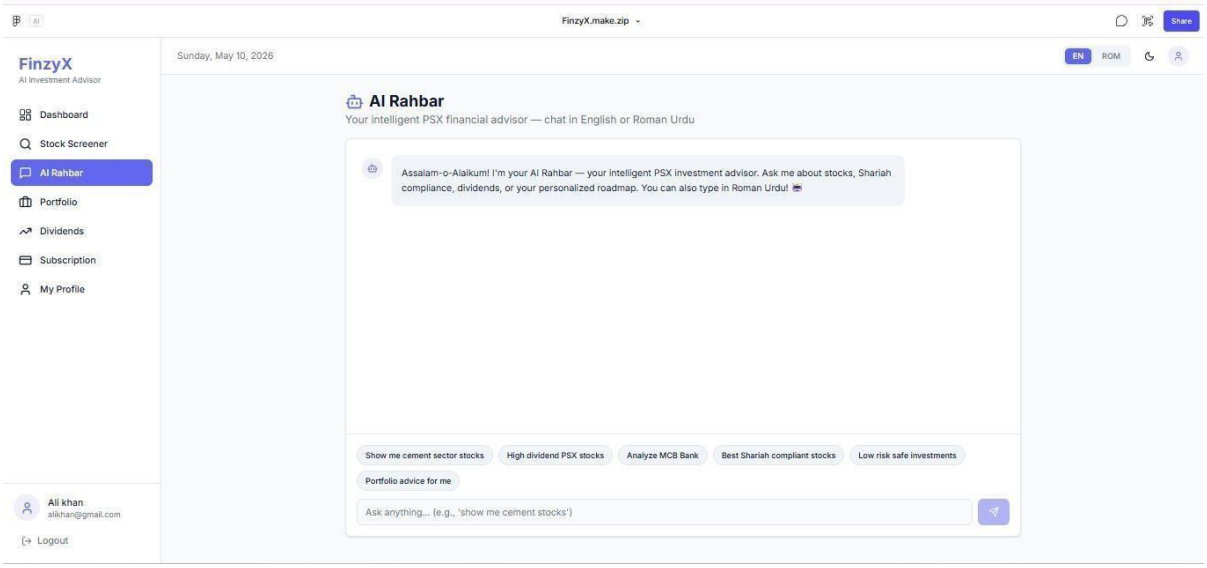


Figure 2-7 AI Rahbar Chat Interface

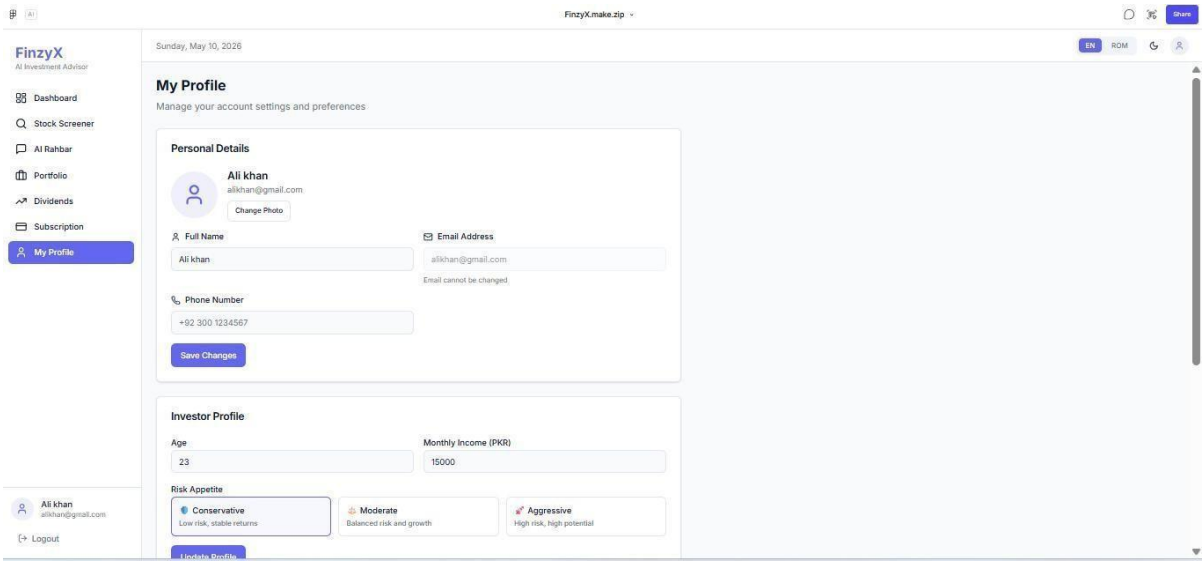


Figure 2-8 Profile Settings Setup

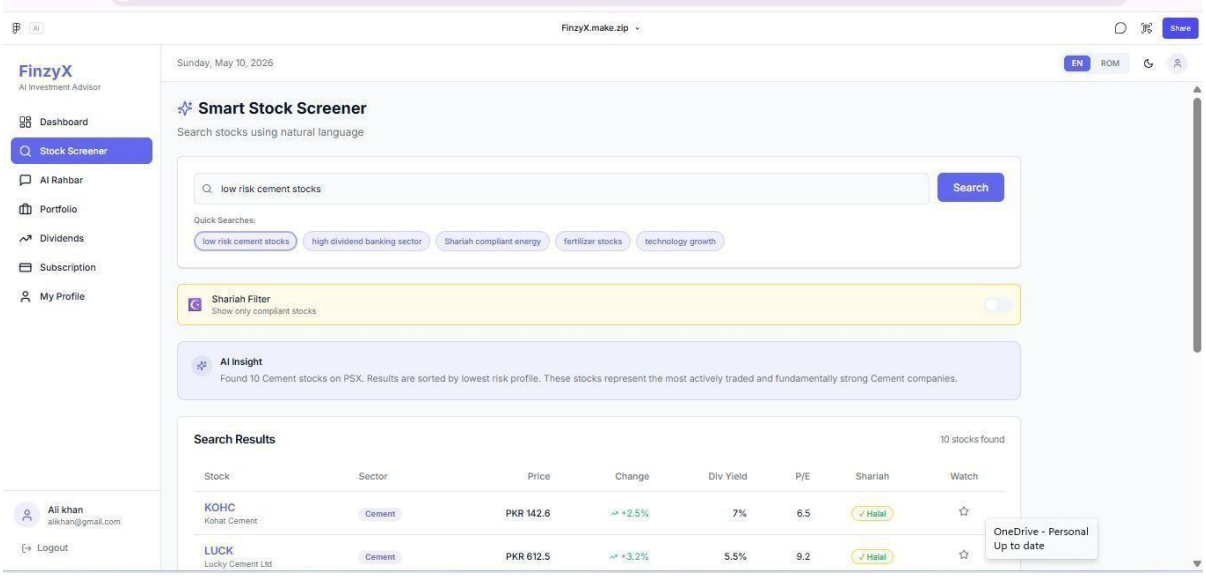


Figure 2-9 Stock Screener Search Bar and Results

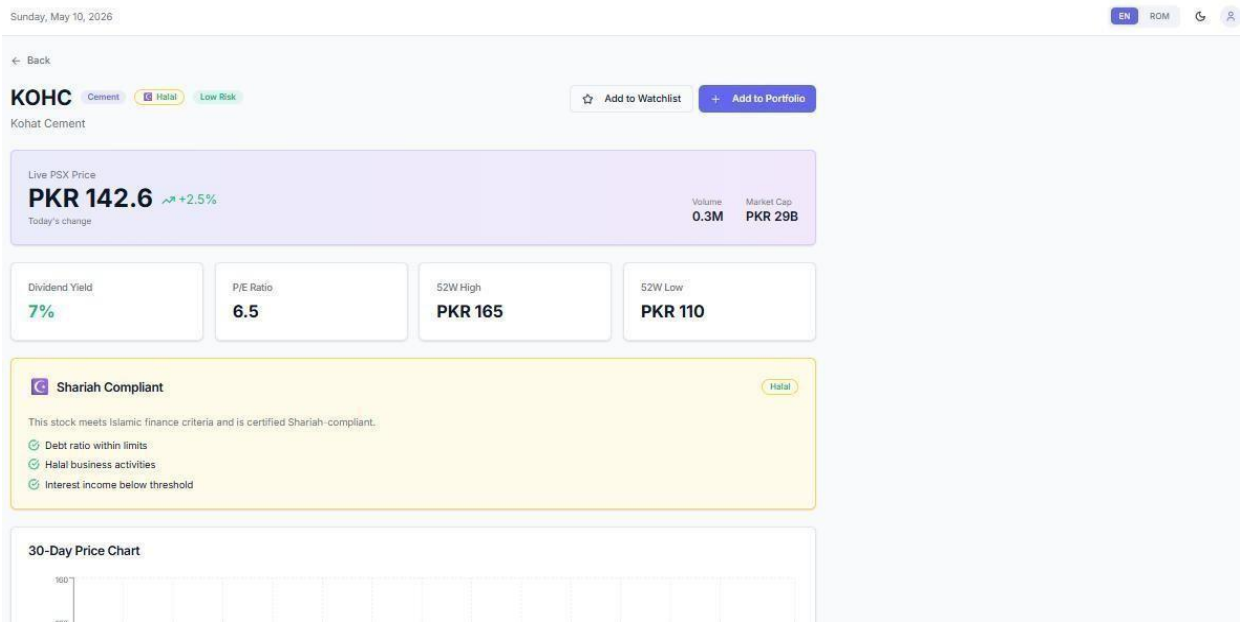


Figure 2-10 Stock Detail Page

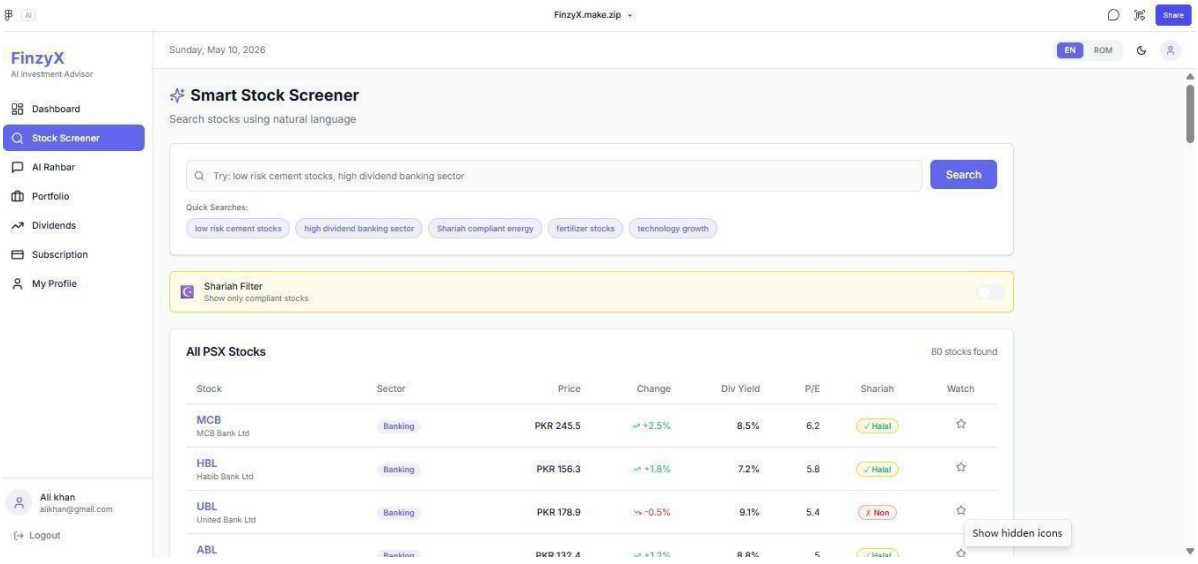


Figure 2-11 Screener Empty State

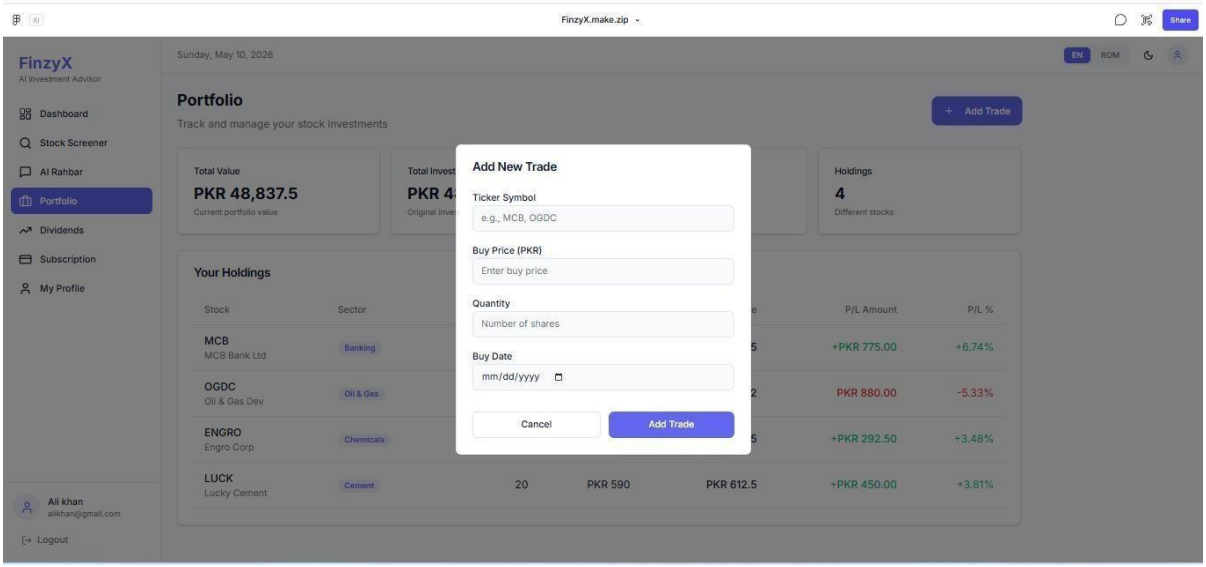


Figure 2-12 Add Trade to Portfolio

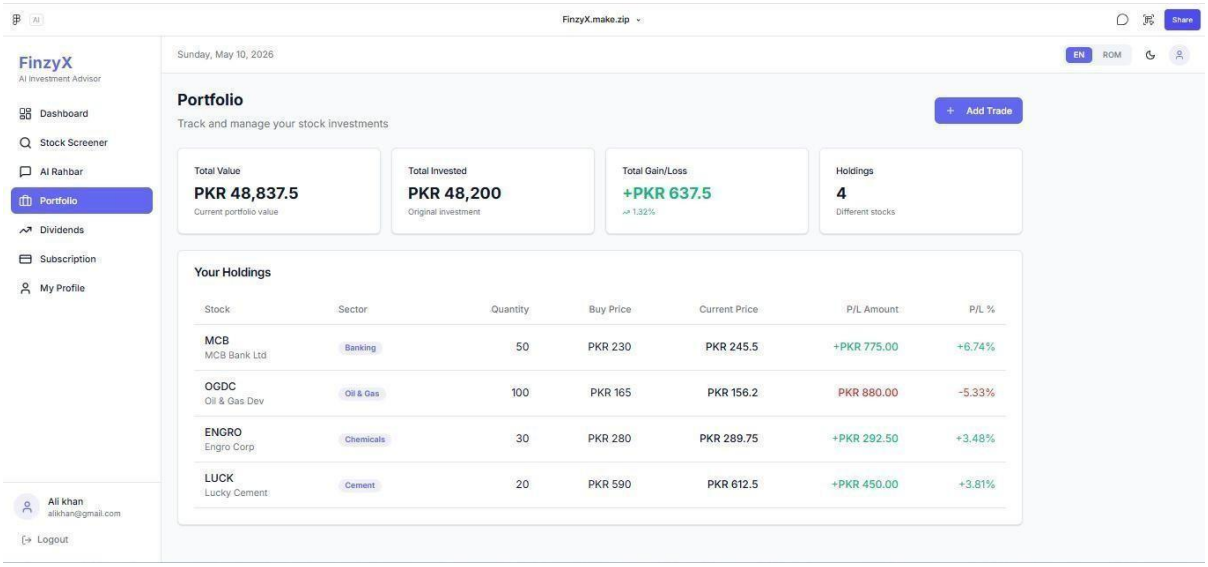


Figure 2-13 Portfolio Dashboard with Luve P&L

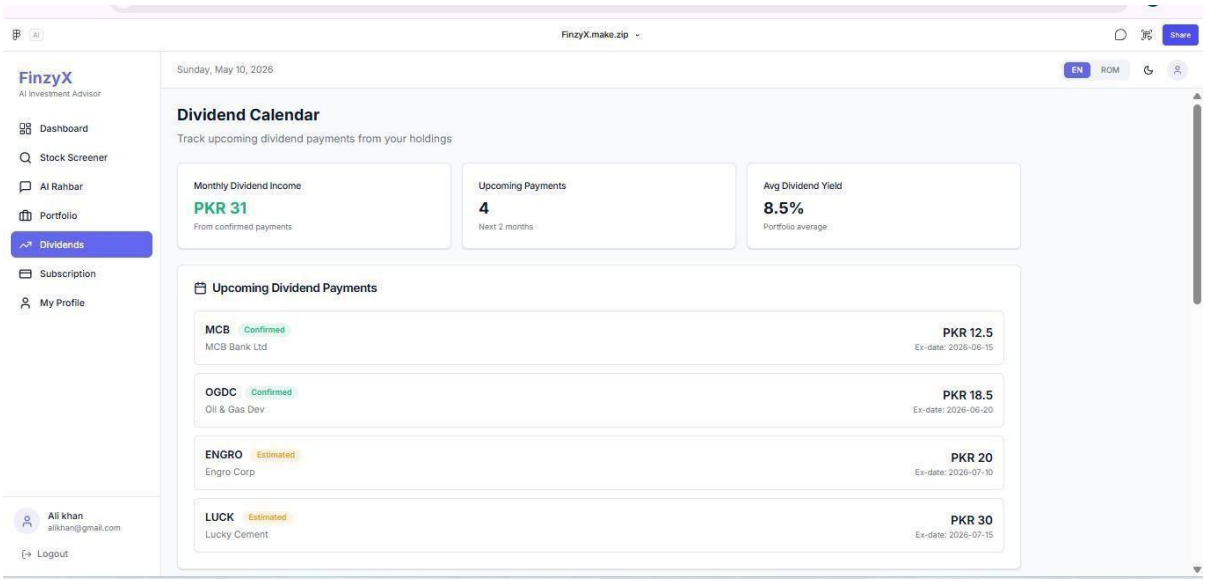


Figure 2-14 Dividend Calendar View

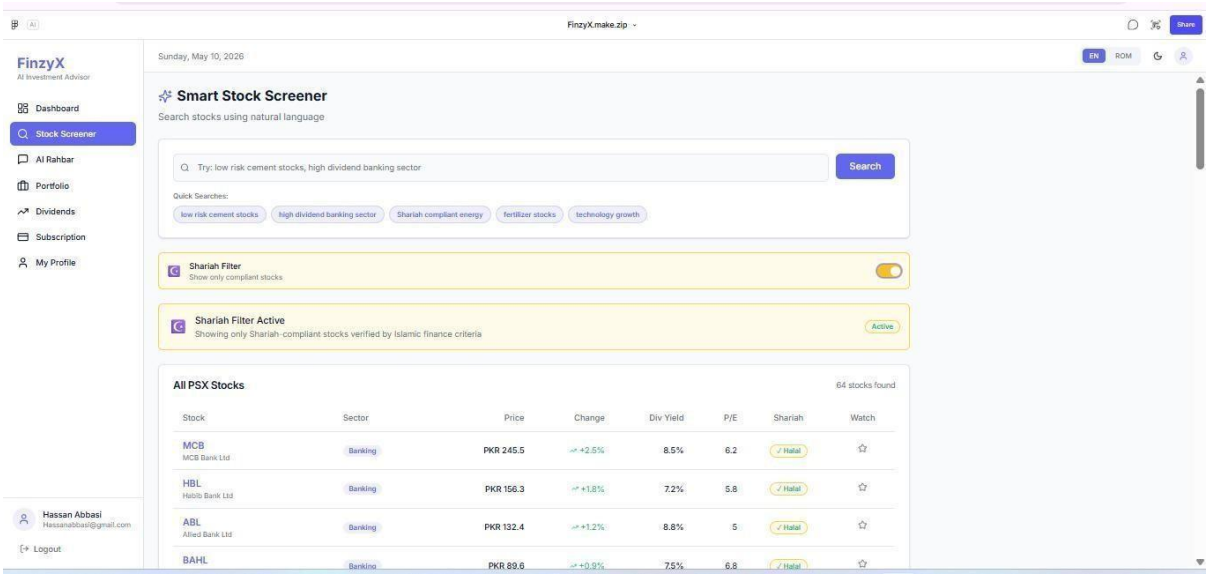


Figure 2-15 Screener with Shariah Filter Enabled

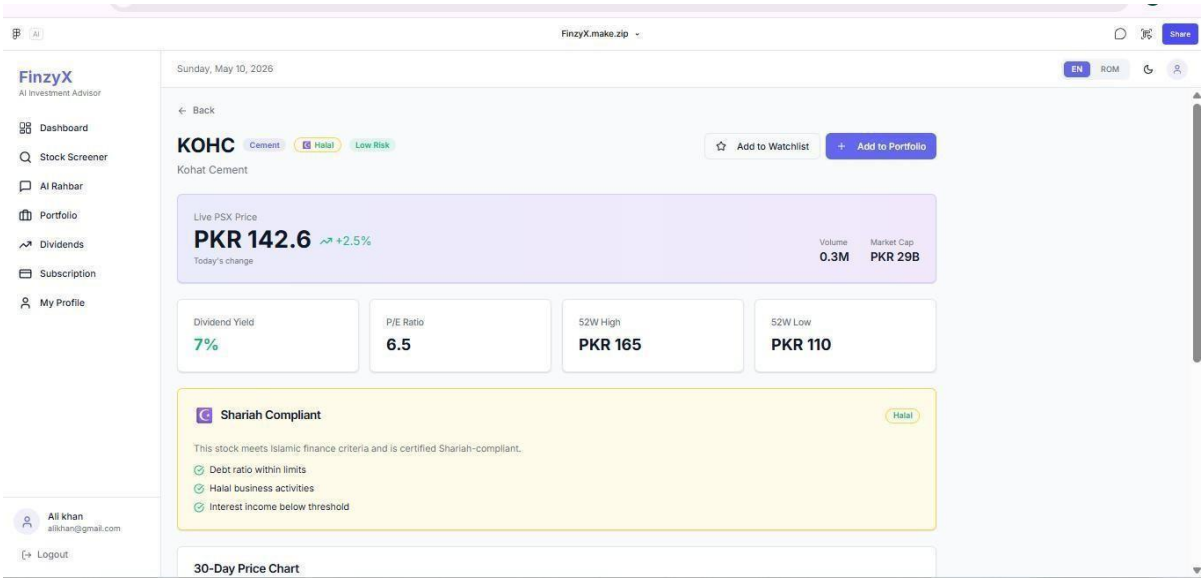


Figure 2-16 Stock Detail Page Shariah Label

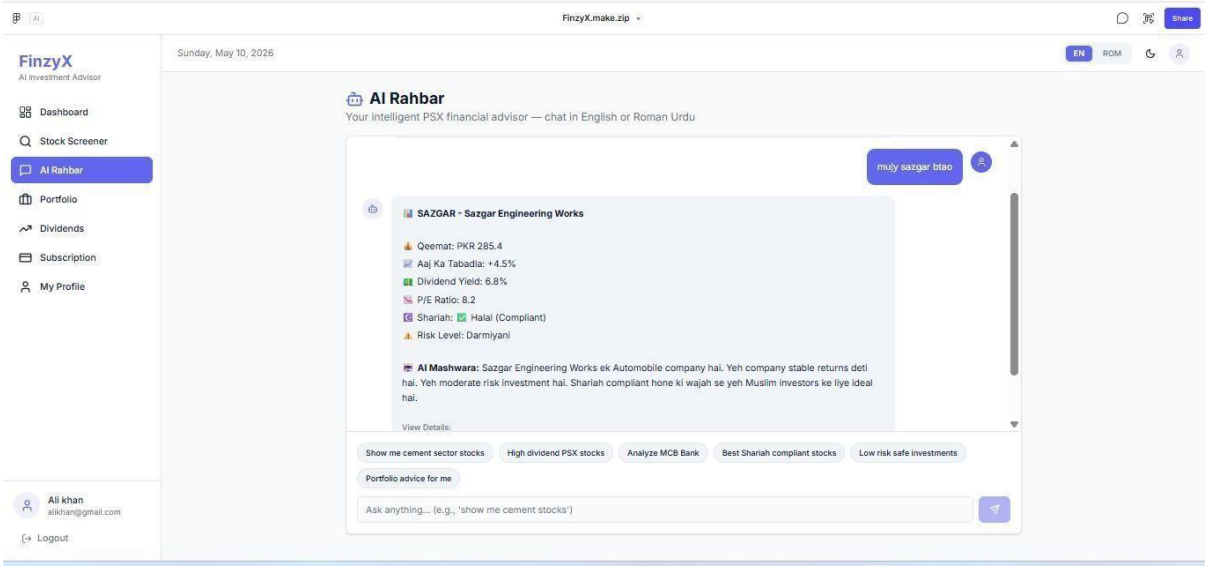


Figure 2-17 AI Rahbar Chat in Roman Urdu

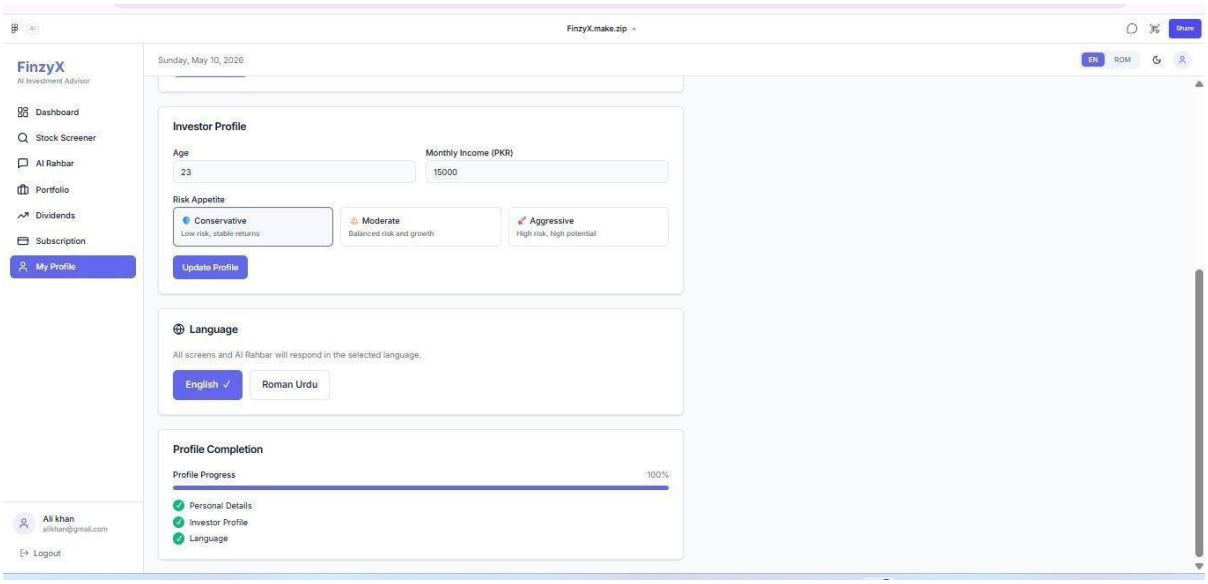


Figure 2-18 Language Preference Settings

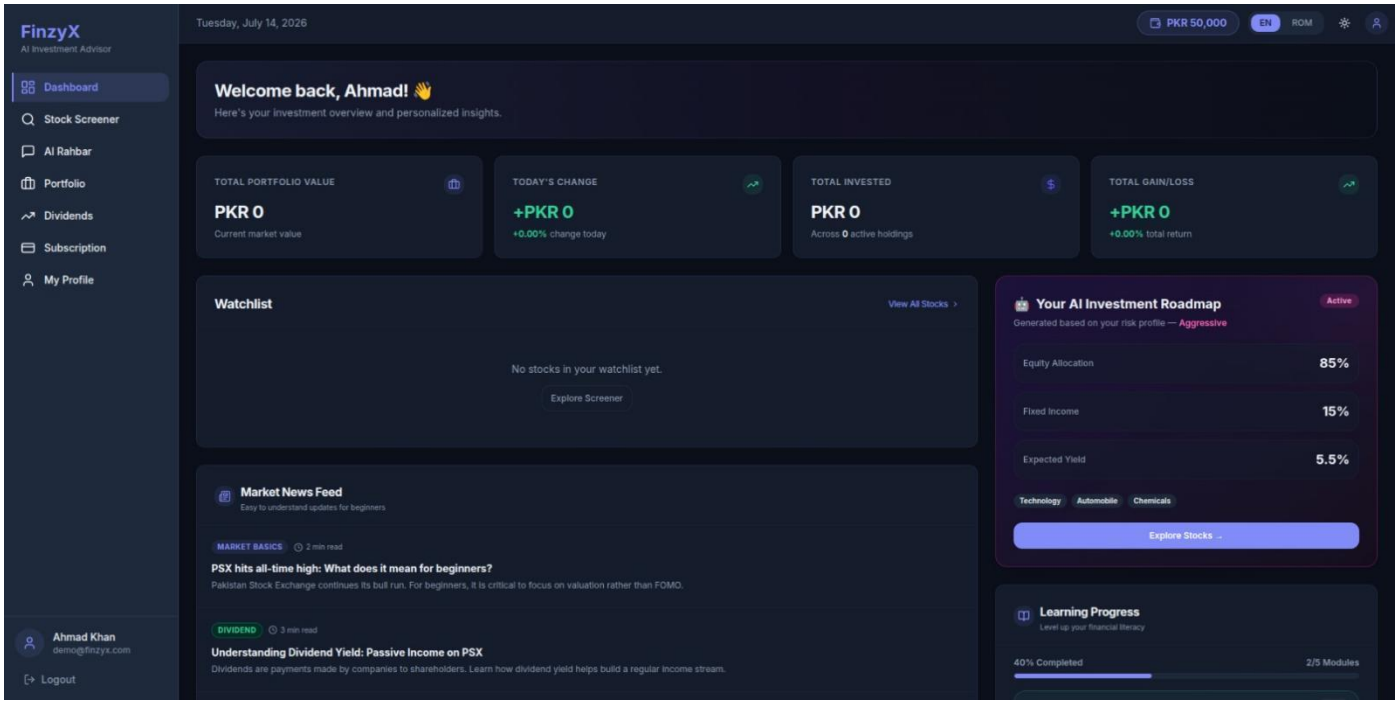


Figure 2-19 Dashboard displaying PKR 50,000 Dummy Balance

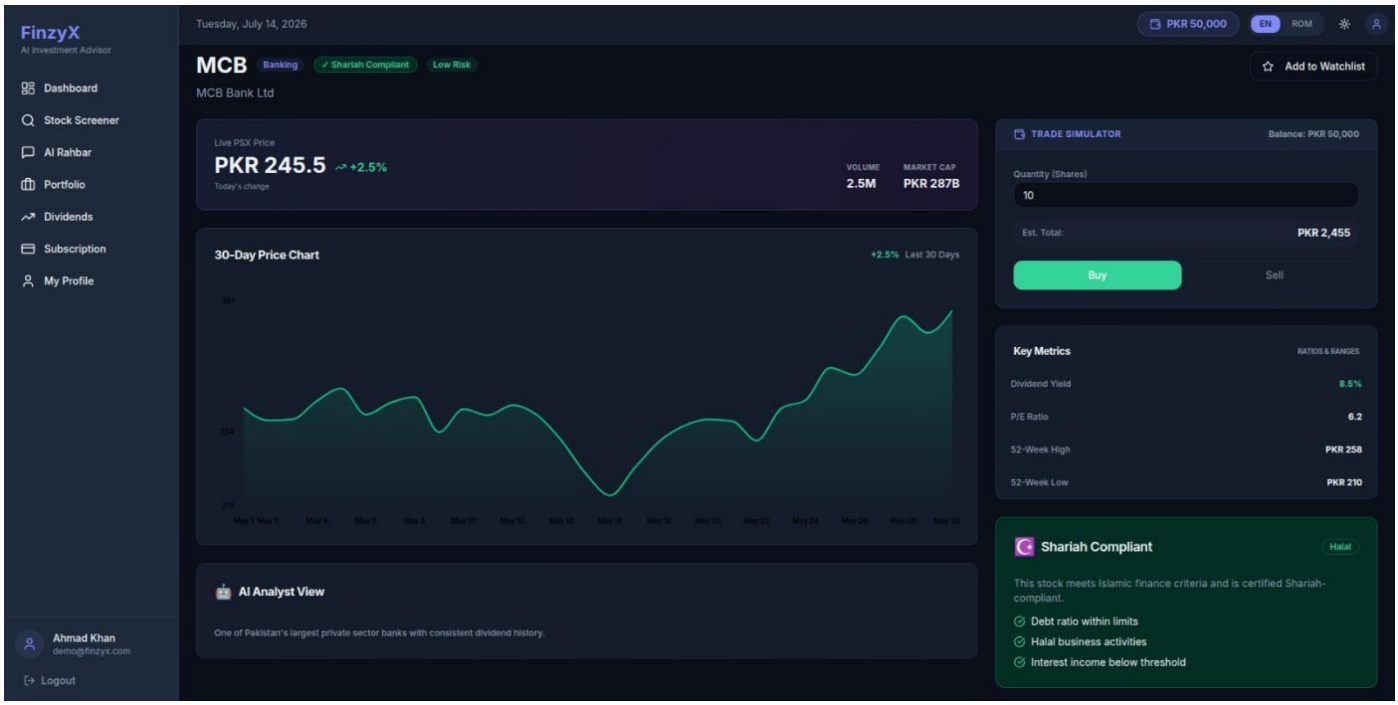


Figure 2-20 Buy or sell

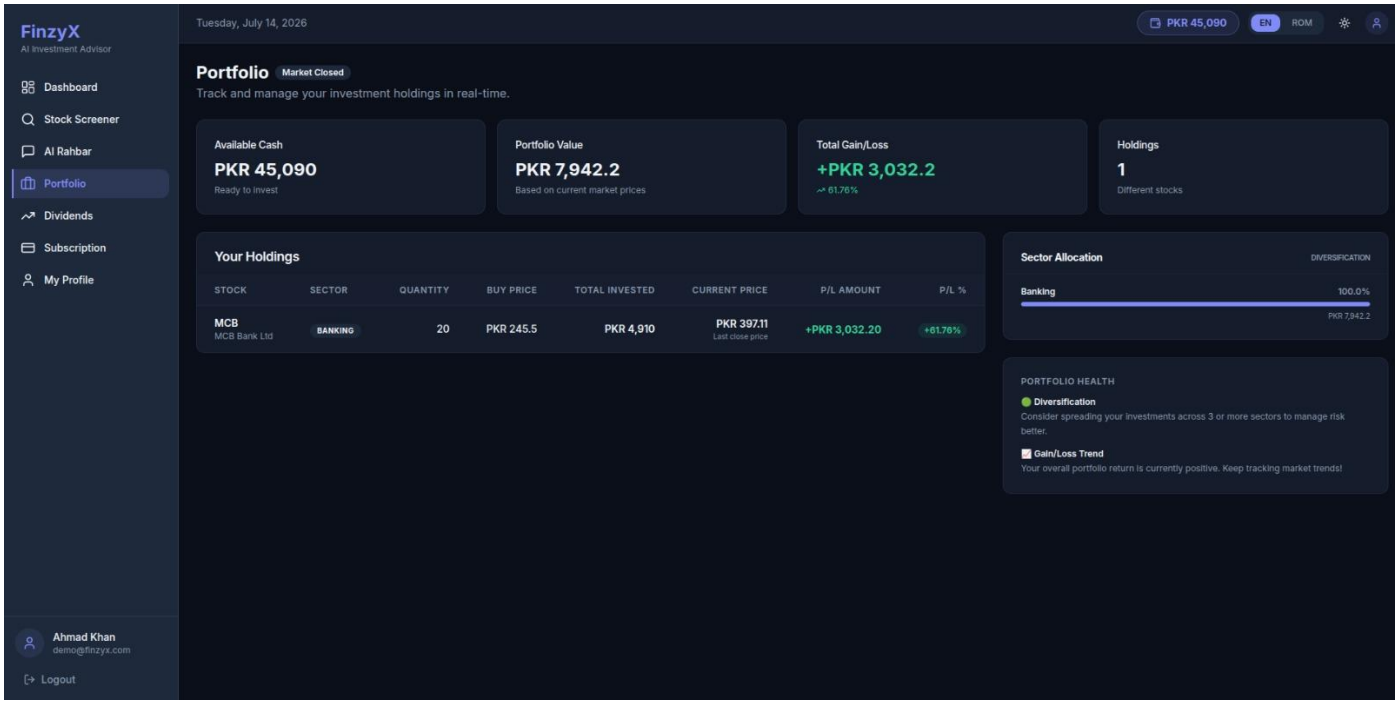


Figure 2-21 Updated Portfolio showing Profit/Loss and Remaining Dummy Balance

2.5 Traceability Matrix

The traceability matrix maps every epic in FinzyX to its user stories, the test cases that validate them, and the UI screens where those features appear. This mapping confirms that every requirement has a corresponding test and a place in the interface. It also makes it straightforward to check for gaps: if a user story has no test case entry, something is missing from the test plan. The UI reference column uses placeholder labels that correspond to the interface screenshots in the following section.

Table 2-28: Requirements Traceability Matrix

Epic	User Story (US)	Test Case (TC)	UI Reference (UI)
E1	E1-US1	E1-US1-TC1	E1-US1-UI1
E1	E1-US1	E1-US1-TC2	E1-US1-UI2
E1	E1-US2	E1-US2-TC1	E1-US2-UI1
E1	E1-US2	E1-US2-TC2	E1-US2-UI2
E1	E1-US2	E1-US2-TC3	E1-US2-UI3
E1	E1-US3	E1-US3-TC1	E1-US3-UI1
E1	E1-US3	E1-US3-TC2	E1-US3-UI2

Epic	User Story (US)	Test Case (TC)	UI Reference (UI)
E1	E1-US4	E1-US4-TC1	E1-US4-UI1
E1	E1-US4	E1-US4-TC2	E1-US4-UI2
E1	E1-US4	E1-US4-TC3	E1-US4-UI3
E2	E2-US1	E2-US1-TC1	E2-US1-UI1
E2	E2-US1	E2-US1-TC2	E2-US1-UI2
E2	E2-US2	E2-US2-TC1	E2-US2-UI1
E2	E2-US2	E2-US2-TC2	E2-US2-UI2
E2	E2-US3	E2-US3-TC1	E2-US3-UI1
E3	E3-US1	E3-US1-TC1	E3-US1-UI1
E3	E3-US1	E3-US1-TC2	E3-US1-UI2
E3	E3-US2	E3-US2-TC1	E3-US2-UI1
E3	E3-US3	E3-US3-TC1	E3-US3-UI1
E4	E4-US1	E4-US1-TC1	E4-US1-UI1
E4	E4-US1	E4-US1-TC2	E4-US1-UI2
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E4	E4-US2	E4-US2-TC2	E4-US2-UI2
E4	E4-US3	E4-US3-TC1	E4-US3-UI1
E4	E4-US3	E4-US3-TC2	E4-US3-UI2
E4	E4-US4	E4-US4-TC1	E4-US4-UI1
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E4	E4-US5	E4-US5-TC1	E4-US5-UI1
E4	E4-US5	E4-US5-TC2	E4-US5-UI2
E5	E5-US1	E5-US1-TC1	E5-US1-UI1
E5	E5-US1	E5-US1-TC2	E5-US1-UI2
E5	E5-US2	E5-US2-TC1	E5-US2-UI1
E5	E5-US2	E5-US2-TC2	E5-US2-UI2
E6	E6-US1	E6-US1-TC1	E6-US1-UI1
E6	E6-US1	E6-US1-TC2	E6-US1-UI2
E6	E6-US2	E6-US2-TC1	E6-US2-UI1

E7	E7-US1	E7-US1-TC1	E7-US1-UI1
E7	E7-US1	E7-US1-TC1	E7-US1-UI2
E7	E7-US2	E7-US2-TC1	E7-US1-UI2
E7	E7-US2	E7-US2-TC2	E7-US2-UI2
E7	E7-US3	E7-US3-TC1	E7-US3-UI1
E7	E7-US3	E7-US3-TC2	E7-US3-UI1

Chapter 3

3. System Design

This chapter presents the architectural design of the proposed system, FinzyX – An AI-Powered Investment Assistant for Beginner Investors in Pakistan, and serves as the bridge between the system requirements identified in Chapter 2 and the actual implementation of the application. It explains how the different components of FinzyX are organized, how they communicate with one another, and how they collectively deliver intelligent financial advisory services to users. The objective of this chapter is to provide a comprehensive design of the proposed system that ensures all functional and non-functional requirements are translated into an efficient, scalable, and maintainable software solution.

The chapter discusses the overall software architecture, database design, workflow of the system, and the third-party technologies integrated into FinzyX. The design emphasizes modularity, scalability, and maintainability by separating the presentation layer, business logic, artificial intelligence components, and data storage. Furthermore, the chapter explains how Retrieval-Augmented Generation (RAG), vector databases, and large language models are integrated to provide personalized financial guidance while maintaining reliable access to Pakistan Stock Exchange (PSX) data.

3.1. Software Architecture

Describes the high-level architecture of FinzyX, which is based on a client–server and cloud-based architecture. It highlights the interaction between the frontend interface, backend services, AI processing modules, databases, and external services such as Google Gemini API, Supabase Authentication, Pinecone Vector Database, and Pakistan Stock Exchange (PSX) data sources.

Figure 3.1 illustrates the overall software architecture of FinzyX and demonstrates the

interaction between users, frontend components, backend services, artificial intelligence modules, databases, and external APIs.

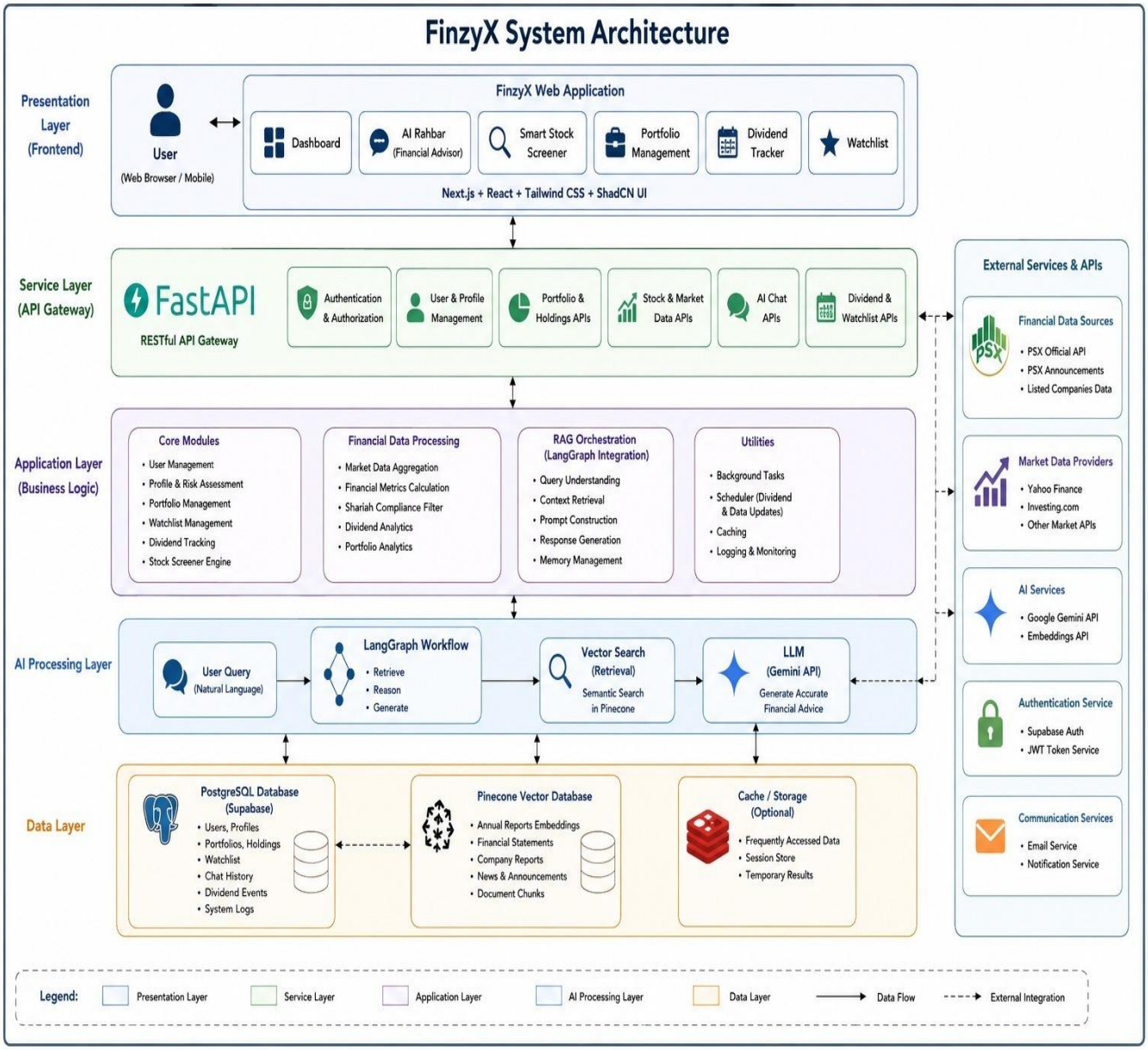


Figure 3-1 System Architecture Diagram

3.2. Data Modeling:

Data modeling defines how information is structured, organized, and managed within the FinzyX system. It ensures that the database design aligns with the functional requirements while supporting efficient storage, retrieval, and management of financial data. Since FinzyX manages structured information such as user accounts, investor profiles, portfolios, stock holdings, watchlists, dividend records, and AI chat history, a well-designed data model is essential to maintain relationships between different system entities. The data model provides a clear representation of how information flows throughout the application and supports scalability as the number of users and financial records increases.

FinzyX also integrates a Financial News Module, which retrieves the latest market news from external financial news APIs. The backend periodically requests updated news articles, processes the received data, and provides users with recent market developments, company announcements, and economic updates through the web application. This feature helps users stay informed about current market conditions and supports better investment decision-making

Depending on the system requirements, FinzyX utilizes a combination of relational and vector-based data storage approaches. The relational database is used to manage structured and transactional data such as user information, portfolios, stock holdings, and watchlists, while the vector database stores financial knowledge embeddings used by the Retrieval-Augmented Generation (RAG) system to provide accurate and context-aware investment recommendations. This hybrid approach improves both data consistency and AI response quality while ensuring efficient system performance.

3.2.1. Relational Databases

A relational database organizes data into structured tables consisting of rows (records) and columns (attributes). In FinzyX, the relational database is used to store and manage structured information such as user accounts, investor profiles, portfolios, stock holdings,

watchlists, dividend records, and chat history. Each table represents a core entity within the system, and relationships between these entities are established using primary and foreign keys to maintain referential integrity and data consistency.

The relational database model is well suited for handling transactional operations, user authentication, and portfolio management, where accuracy and consistency are essential. PostgreSQL, managed through Supabase, is used as the primary relational database because it provides secure data storage, efficient query processing, and high scalability. Using SQL queries, the system can efficiently perform operations such as user registration and login, portfolio management, stock tracking, watchlist updates, and retrieval of historical investment data. This structured database design ensures reliable storage, minimizes data redundancy, and supports the overall functionality of the FinzyX platform.

3.2.2. Entity Relationship Diagram:

The Entity Relationship Diagram (ERD) provides a conceptual representation of the database design for the FinzyX system. It illustrates the primary entities within the application and the relationships among them, enabling developers to understand how data is logically organized before implementation. The ERD serves as a blueprint for the database structure and ensures that all system components are properly connected while minimizing data redundancy.

In the FinzyX ERD, entities such as User, Profile, Stock, Holding, Watchlist, Dividend Event, and Chat Log are represented along with their relationships. A User maintains a single investment Profile that stores personal financial information and investment preferences. Each user can own multiple Holdings, create multiple Watchlist entries, and generate multiple Chat Logs through interactions with the AI Rahbar module. The Stock entity stores information related to Pakistan Stock Exchange (PSX) companies, including company details, sector information, current stock prices, dividend yield, and Shariah compliance status. Each stock can be associated with multiple holdings, watchlist entries,

and dividend events, establishing one-to-many relationships across the database.

The ERD also demonstrates the use of primary keys and foreign keys to maintain referential integrity between related entities. These relationships ensure that user-specific investment data remains consistent while allowing efficient retrieval of portfolios, watchlists, dividend information, and AI interaction history. During the implementation phase, this conceptual data model is transformed into a physical database schema where tables, constraints, and indexes are created within PostgreSQL. The Entity Relationship Diagram provides a strong foundation for implementing the database while ensuring scalability, consistency, and efficient management of financial information.

Figure 3.2 illustrates the Entity Relationship Diagram of the FinzyX system.

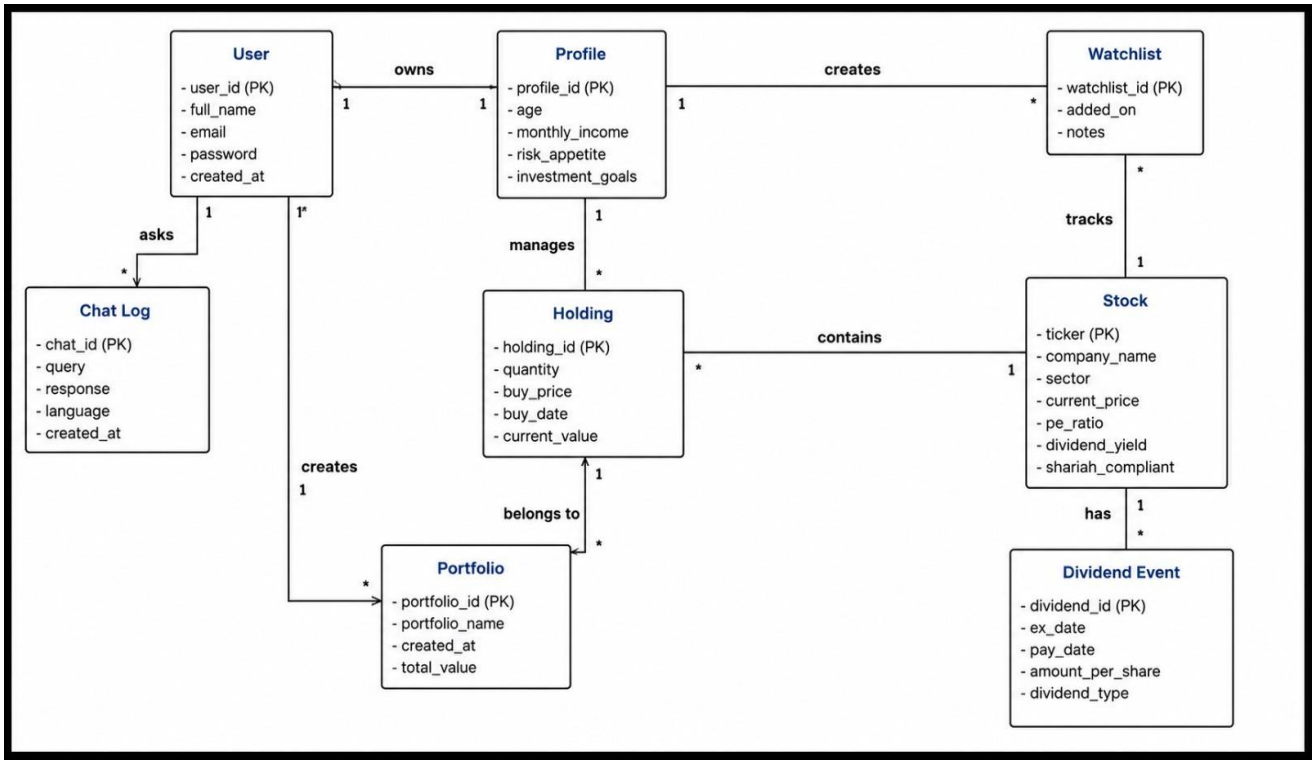


Figure 3-2 Entity Relationship Diagram

3.2.3. Database Schema:

The database schema of FinzyX defines the physical structure of the relational database and illustrates how the system stores and manages financial information. It represents the database tables, their attributes, primary keys, foreign keys, and the relationships established between them. The schema is designed to ensure data integrity, minimize redundancy, and support efficient data retrieval for various system operations.

The FinzyX database consists of several core tables, including Users, Profiles, Stocks, Holdings, Watchlists, Dividend Events, and Chat Logs. Each table stores specific information related to the application's functionality while maintaining relationships through primary and foreign keys. For example, the Users table stores authentication and account information, while the Profiles table maintains investor-specific details such as monthly income, investment goals, and risk tolerance. Similarly, the Holdings and Watchlists tables are linked to both users and stocks, allowing users to manage their investment portfolios and monitor selected stocks efficiently.

The database schema has been developed using a visual database modeling tool to provide a clear representation of the physical database design. Appropriate normalization techniques have been applied to eliminate redundant data and improve consistency across all tables. Primary and foreign key constraints enforce referential integrity, while indexing is applied to frequently accessed attributes such as user_id, ticker, and email to enhance query performance. This schema provides a reliable foundation for implementing backend services, portfolio management, stock analysis, dividend tracking, and AI-assisted financial recommendations within the FinzyX platform.

Figure 3.3 presents the database schema of the FinzyX system.

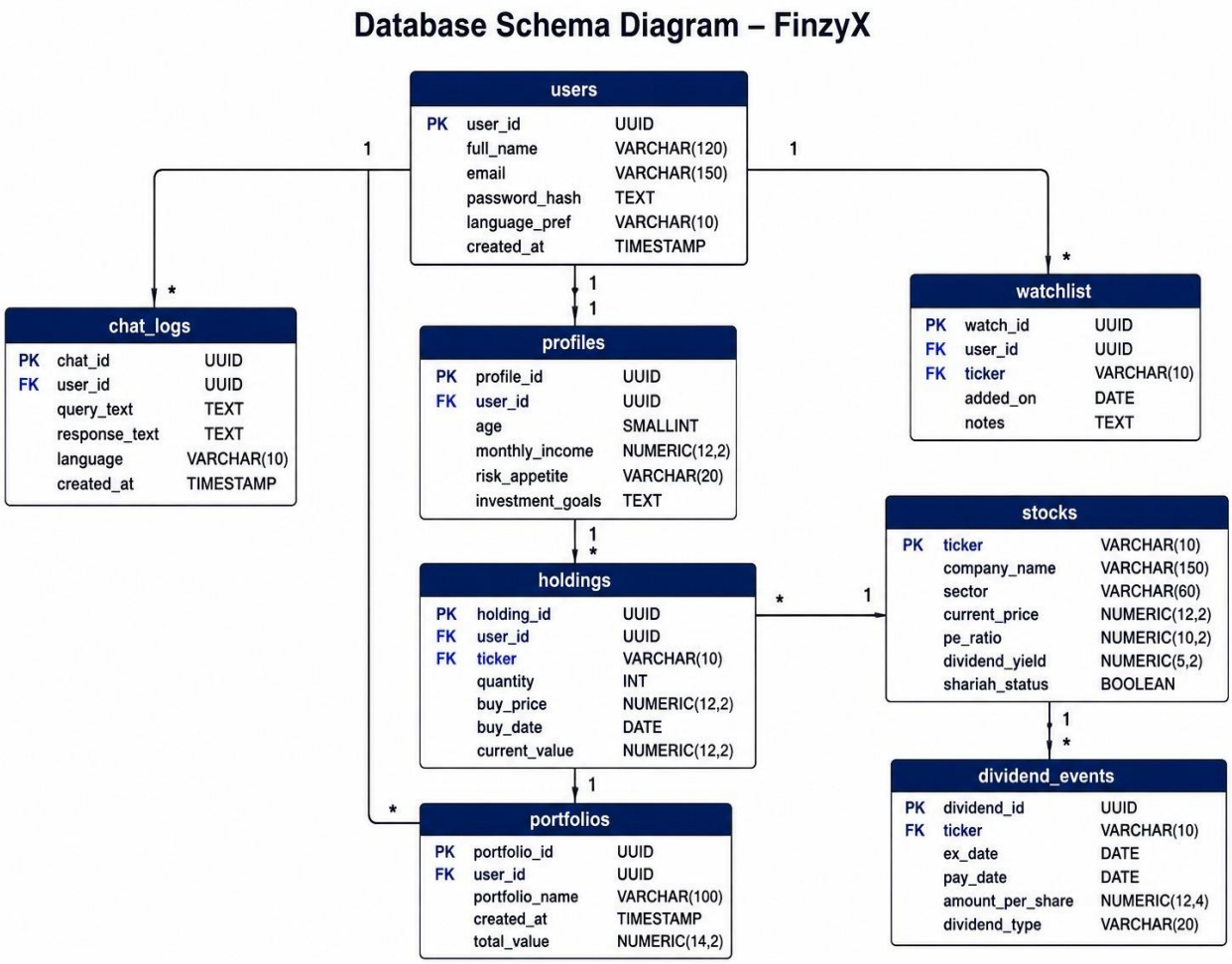


Figure 3-3 Database Schema

3.2.4. Vector Database (Pinecone)

In addition to the relational database, FinzyX utilizes a Pinecone Vector Database to support its Artificial Intelligence (AI) powered financial advisory system. Unlike relational databases that store structured information in tables, a vector database stores high-dimensional embeddings generated from financial documents, investment guides, and Pakistan Stock Exchange (PSX) data. These embeddings allow the system to perform semantic searches by understanding the meaning and context of a user's query rather than relying solely on keyword matching.

The vector database plays a key role in the Retrieval-Augmented Generation (RAG)

architecture implemented in FinzyX. When a user asks a financial question through the AI Rahbar module, the query is first converted into a vector embedding and matched against the stored financial knowledge in Pinecone. The most relevant documents are then retrieved and provided as context to the Google Gemini API, which generates an accurate and context-aware response. This retrieval process significantly improves the quality and reliability of AI-generated financial recommendations.

Using a vector database provides several advantages, including faster semantic search, improved scalability, and better retrieval accuracy for large collections of financial documents. It also allows FinzyX to update its knowledge base without modifying the underlying AI model, making the system more flexible and easier to maintain as new financial data becomes available.


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 "document_id": "DOC_101",
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 "metadata": {
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  "language": "English"
 }
}
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3.3. Workflow Diagram:

A workflow diagram represents the sequence of activities and interactions performed by different components of the system to accomplish a specific task. It provides a visual representation of how information flows between users, the application, artificial intelligence modules, databases, and external services. Workflow diagrams help developers and stakeholders understand the execution process, identify decision points, and ensure that the system operates according to the defined functional requirements.

In FinzyX, the workflow begins when a user accesses the web application and logs into the system using secure authentication. After successful authentication, the user can complete their investment profile, manage their portfolio, search Pakistan Stock Exchange (PSX) stocks, maintain a watchlist, or interact with the AI Rahbar assistant. When a user submits a financial query, the request is forwarded to the backend, where the query is processed and converted into vector embeddings. The system then retrieves the most relevant financial information from the Pinecone Vector Database using the Retrieval-Augmented Generation (RAG) pipeline. The retrieved information is combined with the user's query and passed to the Google Gemini API, which generates a personalized and context-aware financial response. The generated response is then returned to the user through the web interface, while the conversation is securely stored in the database for future reference.

For portfolio-related operations, users can add, update, or remove stock holdings, create personalized watchlists, and monitor dividend information. All portfolio and investment data is managed through the backend and stored in the PostgreSQL database using Supabase. Administrators are responsible for managing stock information, updating financial datasets, and maintaining the knowledge base used by the AI system. This workflow ensures smooth communication between the frontend, backend, artificial intelligence components, and databases while providing users with accurate, personalized, and real-time investment assistance.

Figure 3.4 illustrates the complete workflow of the FinzyX system, showing the interaction between users, the application, the AI processing module, databases, and administrative functions.

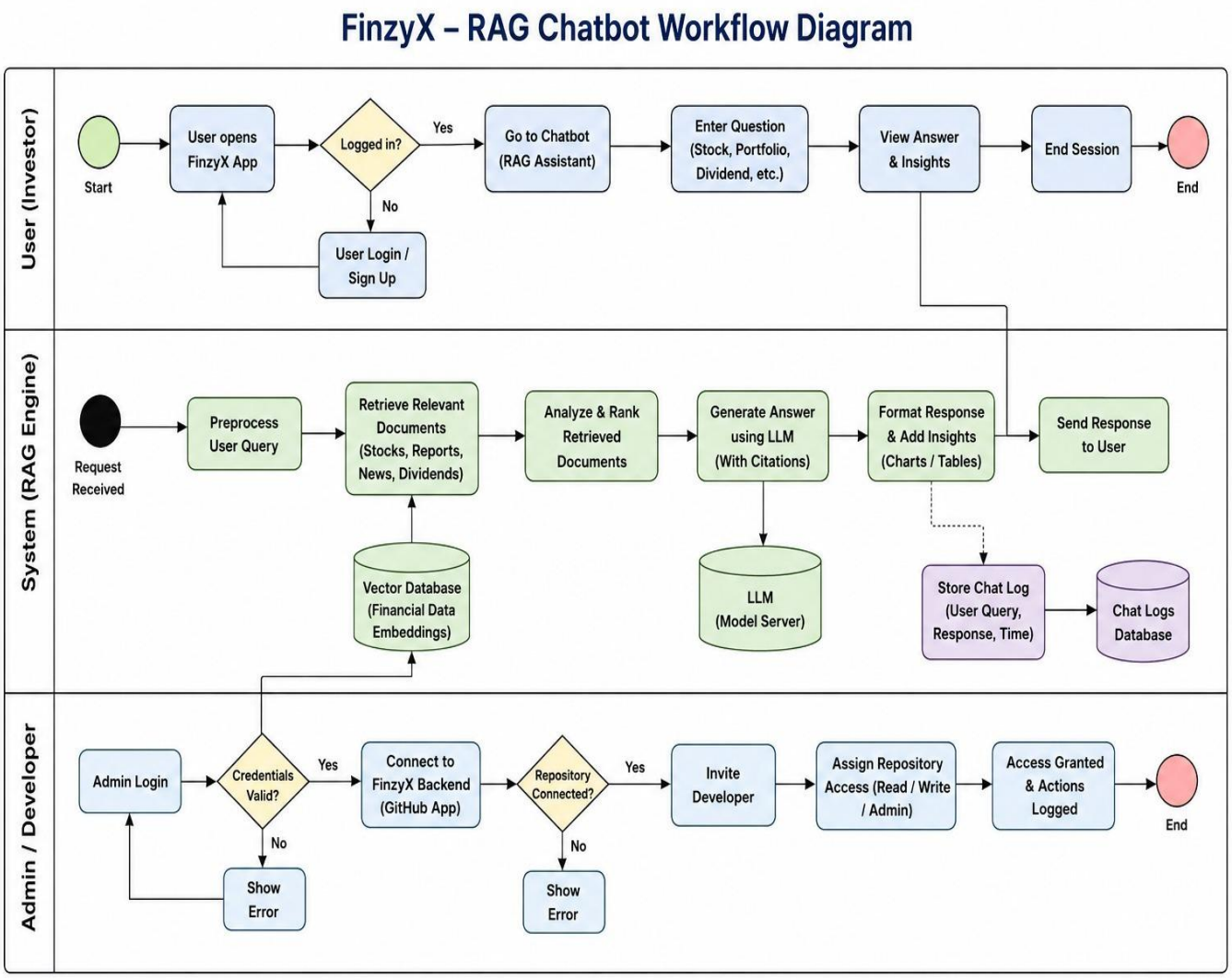


Figure 3-4 Workflow Diagram

3.4. Third-Party Dependencies:

Third-party dependencies refer to external libraries, frameworks, APIs, and cloud services integrated into the FinzyX system to enhance its functionality and simplify the development process. Instead of implementing every feature from scratch, these technologies provide reliable, secure, and well-tested solutions for user authentication, artificial intelligence, database management, financial data processing, and frontend development. Proper documentation of these dependencies ensures maintainability, scalability, and transparency throughout the system design and implementation.

FinzyX relies on several third-party technologies to support its core features, including AI-powered financial guidance, portfolio management, stock analysis, and secure user authentication. These dependencies improve system performance while reducing development complexity and enabling seamless integration between different software components.

The major third-party dependencies used in the FinzyX system are listed below:

- **Next.js:** Used for developing the frontend of the application. It provides server-side rendering, routing, and improved performance while delivering a responsive and user-friendly interface.
- **React.js:** A JavaScript library used to build reusable user interface components and create a dynamic, interactive user experience.
- **Tailwind CSS & Shadcn/UI:** Used to design modern, responsive, and visually consistent user interfaces across different devices.
- **FastAPI:** Serves as the backend framework responsible for processing user requests, implementing business logic, handling REST APIs, and communicating with AI services and databases.
- **Supabase:** Used as the Backend-as-a-Service (BaaS) platform for user authentication, PostgreSQL database management, and secure data storage.
- **PostgreSQL:** The primary relational database used to store structured data such as user accounts, investor profiles, portfolios, stock holdings, watchlists, dividend

information, and chat history.

- **Pinecone Vector Database:** Stores vector embeddings of financial documents and educational content, enabling efficient semantic search within the Retrieval-Augmented Generation (RAG) pipeline.
- **Google Gemini API:** Provides advanced Large Language Model (LLM) capabilities for generating personalized and context-aware financial advice based on retrieved investment knowledge.
- **LangGraph:** Manages the Retrieval-Augmented Generation (RAG) workflow by coordinating query processing, document retrieval, prompt construction, and AI response generation.
- **Pakistan Stock Exchange (PSX) Dataset:** Used as the primary financial data source for stock information, dividend records, company profiles, and market-related information required by the application.

The integration of these technologies enables FinzyX to provide secure authentication, efficient data management, intelligent financial recommendations, and a modern web-based user experience. By leveraging industry-standard frameworks and cloud services, the system achieves high reliability, scalability, and maintainability while supporting future enhancements and additional AI capabilities.

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